

From Forecast to Position: Order-Book Signals, Market Impact, and Causal Trading

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Abstract

An order-book forecast becomes a trading decision through costs, constraints, and the information available when the position is chosen. This paper develops loss identities and certificates for that conversion. Positive curvature yields a convex-conjugate Bregman loss and a quadratic error bound, with a sharp refinement for bounded positions. Purely proportional costs create threshold decisions with linear worst-case sensitivity and sharp margin rates. For scalar forecast intervals, a conjugate-secant formula gives the exact minimax position, including nonsmooth costs and position bounds. Queue depletion attains the corresponding quadratic and linear minimax losses. A finite marked-event book model transfers generator, price-mark, and posterior errors into clock-time forecasts, with individually sharp horizon coefficients. A finite subspace test characterizes features that preserve every conditional-mean price forecast; such features need not define a Markov aggregate. A price-specific reduction residual sharpens the transition-norm comparison. Positive likelihood and future-increment enclosures give an alternative interval certificate before the position is chosen. Information unavailable to the trader has a separate decision cost. An exact convex decomposition separates this observation loss from implementation loss. For delayed book observations, a jump-variance identity prices the unobserved state changes; it distinguishes their first-order latency cost from the smaller bias of failing to propagate a stale state. A two-regime example has an explicit delay beyond which trading optimally stops. For trajectories, an adapted-space residual combines forecast, impact, and optimization errors under convex constraints, with exact affine information decompositions and a verified stationary causal tracking specialization. All results concern declared exogenous book laws and objectives. Synthetic calculations check the identities and rigorous enclosures without empirical profitability claims.

Keywords: forecast evaluation; transaction costs; convex duality; order-book prediction; partial observation; causal trading.

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1 Introduction

An order-book predictor uses queues, trades, cancellations, and waiting times to form a conditional forecast. Its accuracy and trading performance are related through a decision problem. A conditional mean becomes a position; trading costs and constraints determine the conversion; and an implementation must use only information available at the decision time. A forecast can be statistically accurate while its induced position is economically poor, but this does not make statistical consistency irrelevant. The relevant question is which prediction errors matter, with what weights, and under which regularity assumptions.

The fundamental-law literature relates investment performance to signal quality and breadth (Grinold, 1989; Grinold and Kahn, 2000). Gârleanu and Pedersen (2013) derive dynamic portfolios that adjust partially toward an aim incorporating expected future targets and signal decay. More generally, Elmachetoub and Grigas (2022) place the downstream optimization loss at the center of prediction. The present paper uses this decision-based perspective; it does not claim that evaluating a predictor by its decisions is a new principle.

The results are organized around three transfers. The first is from a conditional-mean error to a position loss. Theorem 3.1 expresses that loss as a convex-conjugate Bregman divergence and includes mixed proportional and quadratic frictions. The second is from a finite-state order-book model to the conditional mean: Theorem 5.1 separates generator error from hidden-state posterior error. Theorem 5.11 completes this transfer using marked price events, an event-history posterior, and a residual for state reduction. It exposes the cost of omitting event timing or compressing states with different future book laws. Proposition 5.6 proves that the additive forecast’s horizon and mixing coefficients cannot be reduced uniformly. Theorem 5.16 gives a more detailed alternative: it encloses the event-history likelihood and future price increment before normalization. The resulting scalar interval feeds directly into the exact minimax decision of Theorem 3.4. Example 5.18 compares both certificates under the same primitive uncertainty box. Proposition 5.13 identifies the smaller subspace needed to preserve the price mean at every horizon, and Proposition 5.14 bounds the corresponding observable-specific error. The information decomposition in Theorem 5.19 makes clear when a model certificate is relative to the trader’s information and when a stronger benchmark incurs an additional observation loss. Proposition 5.20 computes that loss for a delayed book feed. The third is from a target position to an implementable trading trajectory. Theorem 6.1 combines errors in the forecast and the impact operator, while Theorem 6.4 separates the cost of restricted information from misspecification within that information class. Proposition 3.3 identifies the sharp uniform envelope when both positive curvature and a finite position diameter are present. Corollary 6.3 identifies the forecast-error directions seen by the implemented trajectory. Theorem 7.1 verifies the causal OU policy through an exact HJB completion-of-squares identity, which also measures the loss of using a wrong tracking rule.

These are tractable conditional models. The convex-duality result uses standard conjugacy (Rockafellar, 1970); the tracking policy is a scalar linear-quadratic control problem. Their use here is to make the loss identities, assumptions, and failure cases explicit in trading units. The finite-state book model is an abstraction of price and queue states, related in spirit to queue-reactive modeling (Huang et al., 2015); it does not prescribe an exchange’s complete order-type or matching system.

Two connections delimit the contribution. Signal-adaptive execution with temporary and transient impact is already developed by Neuman and Voß (2022); strategic interaction between fast and slow traders is studied by Cont et al. (2025). The joint residual below is an error identity on a fixed exogenous probability space, not a new equilibrium or a new solution of those control models. The sharp threshold rates specialize the margin mechanism of Audibert and Tsybakov (2007) to the buy, hold, and sell decision, and explain why global MSE alone and uniformly bounded error lead

to different rates even under the same signal distribution.

Within the trilogy, the execution monograph (Noguer Alonso, 2026) fixes a parent mandate and accounts for feasible instructions and fills. Here the target position is chosen from a predictive signal. The bridge is mathematical: the same joint-residual theorem covers the quadratic execution specialization when the linear signal term is zero, and covers forecast-driven trading when that term is retained. It preserves the declared information class and the common feasible set.

2 The quadratic decision problem

Let $r \in L^2$ be a one-period return, and let $\mathcal{G}$ contain the information available before the position is chosen, including training information if relevant. Put $\mu = \mathbb{E}[r \mid \mathcal{G}]$. Positions and forecasts in this section are $\mathcal{G}$ -measurable and square integrable. For a fixed $a > 0$, define

$$U(w) = \mathbb{E} \left[\mu w - \frac{a}{2} w^2 \right].$$

For example, with constant conditional variance σ^2 , a quadratic risk charge $\gamma\sigma^2 w^2/2$ and quadratic trading cost ηw^2 give $a = \gamma\sigma^2 + 2\eta$. This one-period specification starts from a flat position; it is not a cost on an arbitrary multi-period turnover process.

Theorem 2.1 (Exact squared-error transfer). *The optimal position is $w^* = \mu/a$, with value $\mathbb{E}\mu^2/(2a)$. If a trader uses $\hat{w} = \hat{\mu}/a$, then*

$$U(w^*) - U(\hat{w}) = \frac{\mathbb{E}[(\mu - \hat{\mu})^2]}{2a}. \quad (1)$$

The incremental value of observing $\mathcal{G}$, relative to the best constant position, is $\text{Var}(\mu)/(2a)$.

Proof. The pointwise identity

$$\mu w - \frac{a}{2} w^2 = \frac{\mu^2}{2a} - \frac{a}{2} (w - \mu/a)^2$$

gives the optimizer and (1). The best constant position is $\mathbb{E}r/a$, so subtraction gives the final assertion. $\square$

Corollary 2.2 (Observed-return loss). *For every admissible forecast,*

$$\mathbb{E}[(r - \hat{\mu})^2] = \mathbb{E}[(r - \mu)^2] + \mathbb{E}[(\mu - \hat{\mu})^2].$$

Thus, for fixed information, friction, and sampling law, expected squared error against the observed return ranks forecasts in the same way as the quadratic trading objective.

Proof. The cross term is zero because $\mathbb{E}[r - \mu \mid \mathcal{G}] = 0$. $\square$

For constant a , equal conditional-mean MSE gives equal utility under this fixed plug-in rule. A bias–variance decomposition cannot change that equality. The assertion does not cover leverage constraints, different frictions, or a separately optimized nonlinear calibration. If a is state-dependent, the pointwise identity instead gives $\mathbb{E}[(\mu - \hat{\mu})^2/(2a)]$, provided a is known before trading, bounded away from zero, and the expressions are integrable.

Proposition 2.3 (Multiple assets and a wrong friction matrix). *Let H be symmetric positive definite, and let the conditional objective be $\mu^\top w - \frac{1}{2}w^\top Hw$. For a known H , the plug-in loss is*

$$\frac{1}{2}(\mu - \hat{\mu})^\top H^{-1}(\mu - \hat{\mu}).$$

If a positive definite $\hat{H}$ is also estimated and $\hat{w} = \hat{H}^{-1}\hat{\mu}$, the exact loss is

$$\frac{1}{2}(H\hat{w} - \mu)^\top H^{-1}(H\hat{w} - \mu), \quad H\hat{w} - \mu = (\hat{\mu} - \mu) + (H - \hat{H})\hat{w}. \quad (2)$$

Proof. Complete the square around $H^{-1}\mu$, and use $\hat{H}\hat{w} = \hat{\mu}$. $\square$

The quadratic error metric is therefore set by the portfolio's curvature. Unweighted MSE is exact only in the scalar fixed-curvature case or when the asset metric is a scalar multiple of the identity.

3 Mixed frictions, constraints, and convex duality

Let $F : \mathbb{R}^d \rightarrow (-\infty, \infty]$ be a proper lower semicontinuous friction-and-risk function, including an indicator of a closed convex feasible set when needed. Its convex conjugate is

$$F^*(z) = \sup_w \{\langle z, w \rangle - F(w)\}.$$

Assume F is a -strongly convex for some $a > 0$. This ensures a unique maximizer $w(z)$ for every z . Define the true decision loss from using $\hat{\mu}$ by

$$\mathcal{L}_F(\mu, \hat{\mu}) = F^*(\mu) - \{\langle \mu, w(\hat{\mu}) \rangle - F(w(\hat{\mu}))\}.$$

Theorem 3.1 (Conjugate loss identity and a uniform bound). *Under the stated assumptions, F^* is differentiable, $w(z) = \nabla F^*(z)$, and*

$$\mathcal{L}_F(\mu, \hat{\mu}) = F^*(\mu) - F^*(\hat{\mu}) - \langle \nabla F^*(\hat{\mu}), \mu - \hat{\mu} \rangle \quad (3)$$

$$\leq \frac{\|\mu - \hat{\mu}\|^2}{2a}. \quad (4)$$

The loss is nonnegative. Neither differentiability of F nor interiority of the optimal position is required.

Proof. Strong convexity gives existence and uniqueness of the maximizer and the optimality condition $z \in \partial F(w(z))$. For two forecasts z, z' , strong monotonicity implies

$$a\|w(z) - w(z')\|^2 \leq \langle z - z', w(z) - w(z') \rangle.$$

Thus w is $1/a$ -Lipschitz. Uniqueness of the conjugate maximizer gives differentiability of F^* and $\nabla F^* = w$. Substitution of $F^*(\hat{\mu}) = \langle \hat{\mu}, w(\hat{\mu}) \rangle - F(w(\hat{\mu}))$ proves (3). Convexity gives nonnegativity. Integrating the Lipschitz gradient along the line segment from $\hat{\mu}$ to μ gives (4). $\square$

Corollary 3.2 (Proportional cost with quadratic curvature). *For a scalar position and $F(w) = aw^2/2 + c|w|$, $a, c > 0$,*

$$w(\mu) = \frac{\mathcal{S}_c(\mu)}{a}, \quad \mathcal{S}_c(z) = \text{sgn}(z)(|z| - c)_+, \quad F^*(\mu) = \frac{(|\mu| - c)_+^2}{2a}.$$

With an additional constraint $|w| \leq W$, the solution is clipped to $[-W, W]$. Both models satisfy (4).

Proof. The subgradient condition $\mu \in aw + c\partial|w|$ gives the soft-threshold solution. Substitution gives the conjugate. A scalar concave objective is maximized on a subinterval by clipping its unconstrained maximizer. $\square$

For an existing holding w_0 , the friction

$$F(w) = \frac{a}{2}w^2 + c|w - w_0|$$

has optimizer $w_0 + \mathcal{S}_c(\mu - aw_0)/a$. For multiple assets one may use

$$F(w) = \frac{1}{2}w^\top Hw + \sum_i c_i|w_i - w_{0i}| + I_K(w), \quad H \succeq aI,$$

on any nonempty closed convex set K . These examples show that a proportional component does not by itself destroy the uniform quadratic bound. Positive curvature is the decisive property. As $a \downarrow 0$, the bound degenerates, and the next section describes a different limiting decision problem.

Proposition 3.3 (The sharp curvature–diameter envelope). *For this proposition, first assume only that F is proper, lower semicontinuous, and convex, with finite domain diameter $W = \text{diam}(\text{dom } F)$. For two forecasts with maximizers $w(\mu), w(\hat{\mu})$, put $E = \|\mu - \hat{\mu}\|$. Then*

$$0 \leq \mathcal{L}_F(\mu, \hat{\mu}) \leq \langle \mu - \hat{\mu}, w(\mu) - w(\hat{\mu}) \rangle \leq WE.$$

If F is also a -strongly convex, $a > 0$, then

$$\mathcal{L}_F(\mu, \hat{\mu}) \leq C_{a,W}(E) := \begin{cases} E^2/(2a), & 0 \leq E \leq aW, \\ WE - aW^2/2, & E > aW. \end{cases} \quad (5)$$

For $a, W > 0$, this envelope is attained at every $E \geq 0$ by a one-dimensional constrained quadratic example. In particular it implies $\mathcal{L}_F \leq \min\{E^2/(2a), WE\}$.

Proof. Let $d = w(\mu) - w(\hat{\mu})$. Optimality at $\hat{\mu}$ implies $\hat{\mu} \in \partial F(w(\hat{\mu}))$, hence $F(w(\mu)) - F(w(\hat{\mu})) \geq \langle \hat{\mu}, d \rangle$. Subtracting this inequality from $\langle \mu, d \rangle$ proves the first chain, since $\|d\| \leq W$. Under strong convexity the same subgradient inequality includes $a\|d\|^2/2$. Therefore

$$\mathcal{L}_F(\mu, \hat{\mu}) \leq E\|d\| - \frac{a}{2}\|d\|^2 \leq \max_{0 \leq u \leq W} \{Eu - au^2/2\} = C_{a,W}(E).$$

For sharpness, take $F(w) = aw^2/2 + I_{[-W/2, W/2]}(w)$, where I_C is zero on C and $+\infty$ outside, and choose $\hat{\mu} = -aW/2$, $\mu = \hat{\mu} + E$. The estimated position is $-W/2$; the true position is $\text{clip}_{[-W/2, W/2]}(-W/2 + E/a)$. Direct substitution gives exactly the two branches in (5). If $W = 0$, all decisions coincide and loss is zero. $\square$

The two separate bounds $E^2/(2a)$ and WE intersect at $E = 2aW$, but their minimum is not the sharp combined envelope. Using curvature and diameter in the same inequality gives the transition at $E = aW$ in (5). This is a uniform loss envelope; it does not assert an exact quadratic or linear loss for every forecast pair. For example, $F(w) = aw^2/2 + c|w| + I_{[-W/2, W/2]}(w)$ can have zero loss when both forecasts induce the same no-trade decision. At $a = 0$, the diameter bound remains, while the curvature bound disappears.

Figure 1 shows the decision map and the exact crossover between local curvature and global position diameter.

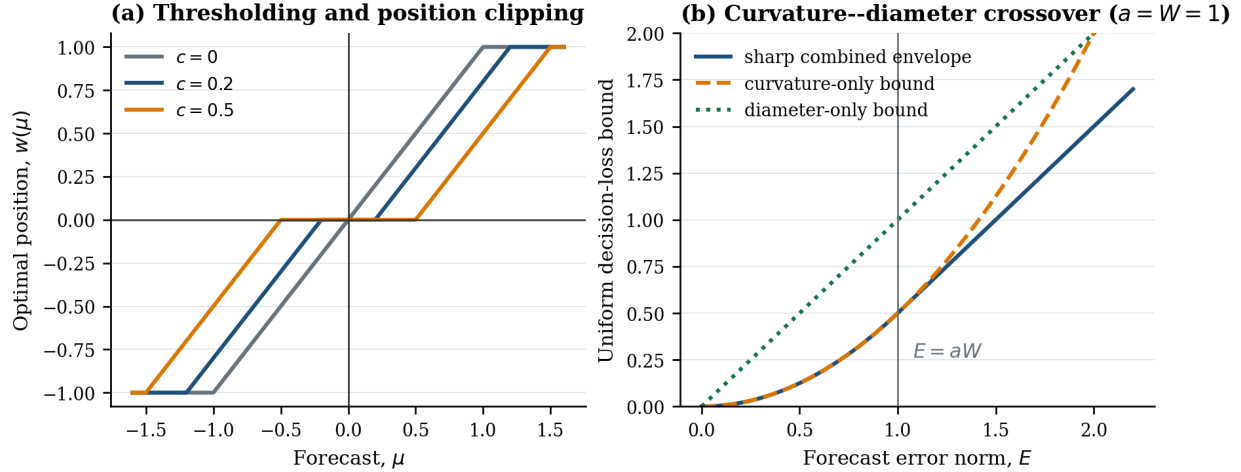


Figure 1: Decision geometry under mixed frictions and constraints. Panel (a) shows scalar soft thresholding with $a = 1$ and clipping to $[-1, 1]$ for three proportional-cost levels. Panel (b) compares the sharp combined envelope $C_{1,1}(E)$ in (5) with the separate curvature-only and diameter-only bounds. The combined inequality changes branch at $E = aW = 1$, before the two separate bounds intersect.

3.1 Exact decisions from a forecast interval

A scalar certificate can sometimes retain the forecast interval itself rather than replace it by a norm bound. For this subsection, assume only that $F : \mathbb{R} \rightarrow (-\infty, \infty]$ is proper, closed, and convex. Let $l < u$, and assume the conjugate values at both endpoints are finite and attained. This includes the strongly convex frictions above and purely proportional costs on a bounded position set.

Theorem 3.4 (The exact interval minimax position). *For a fixed feasible position w , the worst decision loss over $\mu \in [l, u]$ is*

$$\mathcal{R}_{[l,u]}(w) := \sup_{\mu \in [l,u]} \{F^*(\mu) - \mu w + F(w)\} = F(w) + \max\{F^*(l) - lw, F^*(u) - uw\}. \quad (6)$$

An exact minimax position and its loss are

$$w_c = \frac{F^*(u) - F^*(l)}{u - l}, \quad \rho_{[l,u]} = F(w_c) + F^*(l) - lw_c. \quad (7)$$

The position w_c is feasible. If F is strongly convex, it is the unique minimax position. Integrable randomized positions with well-defined expected losses cannot improve on this value.

Proof. For fixed w , the expression being maximized is convex in μ . Its maximum on the interval is therefore attained at an endpoint, proving (6). Let w_l, w_u be conjugate maximizers at the two endpoints. The two supporting-line inequalities for F^* give

$$w_l \leq \frac{F^*(u) - F^*(l)}{u - l} \leq w_u.$$

Both endpoint maximizers belong to $\text{dom } F$; convexity makes every point between them feasible, including w_c .

For $w \leq w_c$, the upper-endpoint term is the larger one, so the risk is $F(w) + F^*(u) - uw$. This convex function is nonincreasing up to any of its minimizers $w_u \geq w_c$. For $w \geq w_c$, the lower-endpoint term is larger and the risk is nondecreasing beyond a minimizer $w_l \leq w_c$. Consequently w_c minimizes the risk and equalizes the two endpoint losses. Strong convexity of F makes the risk strongly convex, giving uniqueness. Finally Jensen's inequality shows that the mean of a randomized position has no larger expected loss at any fixed μ ; taking the worst case proves the last claim. $\square$

For differentiable F^* , this is a one-dimensional convex-conjugate Bregman center. Chebyshev centers for Bregman distances have an established general theory (Bauschke et al., 2010). The scalar calculation above supplies an explicit trading rule, also covering nonsmooth conjugates and position bounds. Its uncertainty set is the whole interval. If the interval is an outer enclosure of a constrained model family, the minimax value for that smaller family can be lower.

Corollary 3.5 (Quadratic and threshold minimax losses). *For $F(w) = aw^2/2$, $a > 0$, on $\mathbb{R}$,*

$$w_c = \frac{l+u}{2a}, \quad \rho_{[l,u]} = \frac{(u-l)^2}{8a}.$$

For $F(w) = c|w| + I_{[-1,1]}(w)$, $c > 0$, and $[l, u] = [c - \epsilon, c + \epsilon]$ with $0 < \epsilon < c$,

$$w_c = \frac{1}{2}, \quad \rho_{[l,u]} = \frac{\epsilon}{2}.$$

If $[l, u] \subseteq \partial F(w_0)$, then $w_c = w_0$ and $\rho_{[l,u]} = 0$.

Proof. The conjugates in the first two cases are $\mu^2/(2a)$ and $(|\mu| - c)_+$, respectively. Substitution in (7) gives the formulas. In the third case, conjugacy gives $F^*(\mu) = \mu w_0 - F(w_0)$ throughout the interval. $\square$

These are matching upper and lower bounds for the stated interval information. In particular, the minimax loss is quadratic in interval width under quadratic friction and linear at a purely proportional trading threshold. The fractional threshold position hedges uncertainty about the forecast. It solves a different decision problem from maximizing profit at one fitted mean.

4 Purely proportional costs and threshold decisions

Now remove the quadratic term and impose $|w| \leq 1$. Let

$$V(w) = \mathbb{E}[\mu w - c|w|], \quad c > 0.$$

Here c is the total proportional charge for the stipulated one-period trade. The feasible bound and absence of quadratic curvature distinguish this model from Section 2.

Proposition 4.1 (Threshold rule and Gaussian value). *An optimal rule is*

$$w^* = \text{sgn}(\mu) \mathbf{1}_{\{|\mu| > c\}}, \quad V(w^*) = \mathbb{E}(|\mu| - c)_+.$$

Ties at $|\mu| = c$ are resolved by taking zero. If $\mu \sim N(0, s^2)$, $s > 0$, then, with $\bar{\Phi}(z) = 1 - \Phi(z)$ the standard normal upper tail,

$$V(w^*) = 2[s\varphi(c/s) - c\bar{\Phi}(c/s)] \sim 2s\varphi(c/s)(s/c)^2, \quad c/s \rightarrow \infty. \quad (8)$$

Proof. On either half of the position interval the objective is linear, so one of $-1, 0, 1$ is optimal. The Gaussian formula follows by integrating $2 \int_c^\infty (x - c)s^{-1}\varphi(x/s) dx$. Using the Mills-ratio expansion $\bar{\Phi}(z) = \varphi(z)(z^{-1} - z^{-3} + O(z^{-5}))$ gives the asymptotic expression. $\square$

The Gaussian value is strictly positive for every $s > 0$. At $c/s = 3$, it is approximately 0.000764309 s ; the zero-cost value is $s\sqrt{2/\pi}$. The retained fraction is approximately 0.09579%. This is a small positive oracle value, not a claim that every weak predictor necessarily produces a loss.

Theorem 4.2 (Exact threshold loss and its worst-case rate). *Let $\hat{w} = \text{sgn}(\hat{\mu})\mathbf{1}_{\{|\hat{\mu}| > c\}}$ and $\varepsilon = \hat{\mu} - \mu$. The pointwise regret is*

$$L = (|\mu| - c)\mathbf{1}_{\{w^* \neq 0, \hat{w}=0\}} + (c - \hat{w}\mu)\mathbf{1}_{\{w^*=0, \hat{w} \neq 0\}} + 2|\mu|\mathbf{1}_{\{\hat{w}=-w^* \neq 0\}}. \quad (9)$$

It satisfies $0 \leq L \leq 2|\varepsilon|$, so $\mathbb{E}L \leq 2\sqrt{\mathbb{E}\varepsilon^2}$. There is no constant C , independent of the joint law, such that $\mathbb{E}L \leq C\mathbb{E}\varepsilon^2$ for every such problem.

Proof. Enumerate the cases where both decisions agree, where one is zero, and where they have opposite nonzero signs to obtain (9). The optimality of $\hat{w}$ for $\hat{\mu}$ gives

$$L \leq (\mu - \hat{\mu})(w^* - \hat{w}) \leq 2|\varepsilon|.$$

For the final assertion take $0 < h < c$, $0 < u < h$, $\mu = c + h$, and $\hat{\mu} = c - u$. Then $L = h$ while $\varepsilon^2 = (h + u)^2$. Letting $u \downarrow 0$ and then $h \downarrow 0$ makes the ratio unbounded. $\square$

The counterexample concentrates mass *near* a moving decision boundary as the error shrinks. It is a worst-case obstruction, not a universal first-order rate for a fixed smooth distribution. The bound $L \leq 2|\varepsilon|$ also shows that MSE consistency implies decision consistency in this bounded-position model.

Proposition 4.3 (A smooth-law expansion). *Suppose μ has a density f continuous and bounded on neighborhoods of c and $-c$. Let a family of errors ε_s , independent of μ , satisfy $\mathbb{E}\varepsilon_s^2 = s^2 \rightarrow 0$ and $\mathbb{E}|\varepsilon_s|^3 = O(s^3)$. Then*

$$\mathbb{E}L = \frac{1}{2}[f(c) + f(-c)]s^2 + o(s^2). \quad (10)$$

Centering the errors is not required.

Proof. Choose $d > 0$ smaller than c and the radii of the density neighborhoods. On $\{|\varepsilon_s| \leq d\}$, opposite nonzero positions cannot occur. Conditional on an error e , the contribution from crossing $+c$ is an integral of $hf(c \pm h)$ over $0 < h < |e|$, with the sign selected by the sign of e . Continuity makes this $\frac{1}{2}f(c)e^2 + o(e^2)$, uniformly as $e \rightarrow 0$. The same argument applies at $-c$. For larger errors,

$$\mathbb{E}[L\mathbf{1}_{\{|\varepsilon_s| > d\}}] \leq 2\mathbb{E}[|\varepsilon_s|\mathbf{1}_{\{|\varepsilon_s| > d\}}] \leq \frac{2}{d^2}\mathbb{E}|\varepsilon_s|^3 = O(s^3).$$

Likewise $\mathbb{E}[\varepsilon_s^2\mathbf{1}_{\{|\varepsilon_s| > d\}}] \leq d^{-1}\mathbb{E}|\varepsilon_s|^3 = O(s^3)$. Choosing a small fixed d , then using uniform continuity on the remaining neighborhood, establishes the averaged $o(s^2)$ remainder. $\square$

Under the independence assumptions and a fixed f , the leading coefficient is the same for all such small error families. With errors correlated with the signal, the allocation of error near the thresholds matters as well. A decision-focused loss can distinguish forecasts with equal global MSE, but the threshold model does not invalidate squared-error estimation of a conditional mean.

4.1 Margin rates without independent errors

The smooth expansion above uses independence. Without it, a small global MSE can concentrate precisely where trading changes sign or switches on. Let

$$D = ||\mu| - c|$$

be distance to the nearest trading threshold. Assume a margin bound

$$\mathbb{P}(D \leq z) \leq Cz^\alpha, \quad 0 < z \leq z_0, \quad C > 0, \quad \alpha > 0. \quad (11)$$

It excludes mass exactly at the thresholds but allows arbitrary dependence between μ and its estimation error.

Theorem 4.4 (Threshold rates under a margin condition). *For the bounded-position proportional-cost decision and (11), the following bounds hold. If $|\varepsilon| \leq e \leq z_0$ almost surely, then*

$$\mathbb{E}L \leq 2Ce^{\alpha+1}.$$

If only $\mathbb{E}\varepsilon^2 \leq s^2$ is known, then for every $0 < z \leq z_0$,

$$\mathbb{E}L \leq 2Cz^{\alpha+1} + \frac{2s^2}{z}. \quad (12)$$

Consequently, for sufficiently small s , $\mathbb{E}L = O((s^2)^{(\alpha+1)/(\alpha+2)})$. For every $\alpha > 0$, this exponent cannot be increased uniformly under only the stated MSE and margin assumptions, even with a fixed law of μ .

Proof. Whenever the true and estimated decisions differ, the interval between μ and $\hat{\mu}$ crosses at least one threshold. Thus $D \leq |\varepsilon|$ on the event of positive loss, and Theorem 4.2 strengthens to

$$L \leq 2|\varepsilon|\mathbf{1}_{\{D \leq |\varepsilon|\}}.$$

The uniform-error bound follows directly. Split the expectation at $|\varepsilon| = z$. On the lower part use $2z\mathbf{1}_{\{D \leq z\}}$; on the upper part use $2|\varepsilon| \leq 2\varepsilon^2/z$. This proves (12). The choice $z = [s^2/(C(\alpha+1))]^{1/(\alpha+2)}$ gives the rate.

For sharpness, fix $0 < d_0 < c$ and a random variable D with $\mathbb{P}(D \leq d) = (d/d_0)^\alpha$ on $[0, d_0]$. Set $\mu = c + D$. For $0 < t < d_0$, define $\varepsilon_t = -2D\mathbf{1}_{\{D \leq t\}}$. The estimated position is zero when $D \leq t$ and agrees with the true unit long position otherwise. Direct integration gives

$$\mathbb{E}L_t = \frac{\alpha t^{\alpha+1}}{(\alpha+1)d_0^\alpha}, \quad \mathbb{E}\varepsilon_t^2 = \frac{4\alpha t^{\alpha+2}}{(\alpha+2)d_0^\alpha}. \quad (13)$$

Eliminating t gives a positive constant times $(\mathbb{E}\varepsilon_t^2)^{(\alpha+1)/(\alpha+2)}$. The signal law is unchanged as $t \downarrow 0$, proving sharpness. $\square$

For a bounded positive density near the thresholds, $\alpha = 1$ is available. An almost-sure error e then gives $O(e^2)$; an MSE budget alone gives the weaker sharp rate $O((\text{MSE})^{2/3})$. The independent small-error expansion of Proposition 4.3 gives $O(\text{MSE})$ because it rules out the concentration mechanism in (13). These are distinct assumptions, not inconsistent rates.

Figure 2 separates the small Gaussian oracle value from the sharp decision-loss rate under dependent forecast errors.

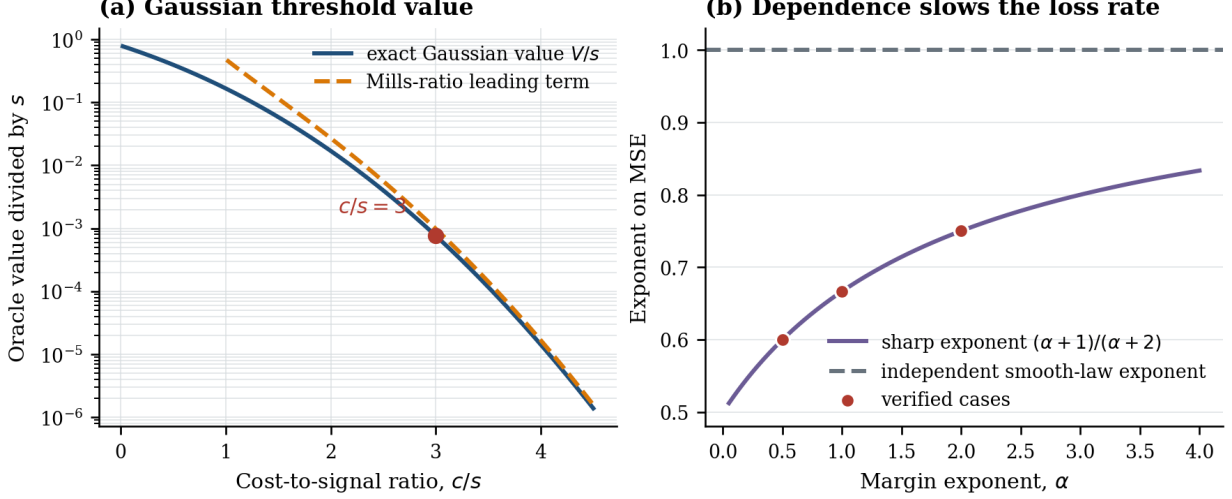


Figure 2: Threshold value and threshold-loss rates. Panel (a) plots the exact normalized Gaussian oracle value from (8) and its Mills-ratio leading term; the marked point is $c/s = 3$. Panel (b) plots the sharp exponent $(\alpha + 1)/(\alpha + 2)$ under only an MSE budget and the margin condition, together with the verified fixed-law cases. The dashed exponent one is the independent smooth-law benchmark of Proposition 4.3; it relies on different assumptions.

5 From order-book events and observations to a trading decision

5.1 Observable and hidden states

Let X_t be a time-homogeneous Markov chain on a finite state space $\{1, \dots, n\}$, with generator Q , nonnegative off-diagonal entries, and zero row sums. Its transition matrix is $P_h = e^{hQ}$. A bounded vector $p \in \mathbb{R}^n$ assigns a reference price to each state. The state may combine displayed queues, spread, recent flow, and a latent liquidity regime. If a quantity is not observed, the trader must use a posterior distribution rather than treat it as visible. Write

$$\pi_t(i) = \mathbb{P}(X_t = i \mid \mathcal{G}_t)$$

for the row-vector posterior given pre-trade observations. Assume the current reference price $p_t = p(X_t)$ is observed and that the model's Markov property holds for the full state. The horizon- h conditional mean price change is then

$$\mu_h(t) = \pi_t e^{hQ} p - p_t. \quad (14)$$

The Markov property here includes the requirement that the observation history contains no additional future information once X_t is given.

This is a local finite-state abstraction. A bounded price grid requires an explicit boundary convention. For real books, time-of-day effects, unmodeled order attributes, and hidden liquidity can invalidate the state reduction. The estimates below are conditional on a specified model and quantify perturbation within that model.

5.2 A generator-to-regret certificate

Theorem 5.1 (Order-book model error transfer). *Let $\hat{Q}$ also be a valid generator on the same state space, and let $\hat{\pi}_t$ be a probability vector. Use the same observed current price and price vector to*

form $\hat{\mu}_h(t) = \hat{\pi}_t e^{h\hat{Q}} p - p_t$. Set

$$\delta_Q = \|\hat{Q} - Q\|_{\infty \rightarrow \infty} = \max_i \sum_j |\hat{Q}_{ij} - Q_{ij}|, \quad B_p = \frac{1}{2}(\max_i p_i - \min_i p_i).$$

Then

$$|\hat{\mu}_h(t) - \mu_h(t)| \leq B_p(\|\hat{\pi}_t - \pi_t\|_1 + h\delta_Q). \quad (15)$$

For an a -strongly convex trading friction,

$$\mathcal{L}_F(\mu_h, \hat{\mu}_h) \leq \frac{B_p^2}{2a}(\|\hat{\pi}_t - \pi_t\|_1 + h\delta_Q)^2. \quad (16)$$

Proof. Stochastic matrices contract the sup norm. Duhamel's formula gives

$$e^{h\hat{Q}} - e^{hQ} = \int_0^h e^{s\hat{Q}}(\hat{Q} - Q)e^{(h-s)Q} ds,$$

hence the operator norm of this difference is at most $h\delta_Q$. Replace p by $p - \frac{1}{2}(\max p + \min p)\mathbf{1}$, whose sup norm is B_p . Constants are annihilated by the difference of probability vectors and by the difference of stochastic matrices. Splitting

$$\hat{\pi}_t e^{h\hat{Q}} - \pi_t e^{hQ} = (\hat{\pi}_t - \pi_t)e^{hQ} + \hat{\pi}_t(e^{h\hat{Q}} - e^{hQ})$$

proves (15). Apply Theorem 3.1 to obtain (16). $\square$

Corollary 5.2 (Estimation rates and the curvature boundary). *For fixed horizon, grid, and positive curvature modulus, if the posterior error and generator error are both $O_p(n_{\text{obs}}^{-1/2})$, the regret bound is $O_p(n_{\text{obs}}^{-1})$. Under the purely proportional bounded-position model, Theorem 4.2 instead gives the general bound*

$$L \leq 2B_p(\|\hat{\pi}_t - \pi_t\|_1 + h\delta_Q).$$

A faster rate there needs additional assumptions about the probability mass and estimation errors near the trading thresholds.

The posterior error includes inference about latent liquidity. An observed queue vector cannot be silently substituted for the full state when hidden variables affect transitions. The theorem does not provide a posterior estimator or establish identifiability. A fitted matrix with negative off-diagonal entries is not a generator and must be repaired before the contraction argument can be used.

Example 5.3 (Two-state calculation). Let $p = (0, 1)^\top$ and

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \lambda & -\lambda \end{pmatrix}.$$

If state zero is observed, then $\mu_h = (1 - e^{-2\lambda h})/2$. For $h = 1, \lambda = 0.4, \hat{\lambda} = 0.6$, the true and estimated means are approximately 0.275336 and 0.349403. Here $B_p = 1/2, \delta_Q = 0.4$, and the forecast bound is 0.2, compared with actual error approximately 0.074067. With quadratic curvature $a = 2$, the exact decision loss is approximately 0.00137149, below the certified upper bound 0.01. All prices and costs in this example are synthetic model units.

5.3 Relative-book states, price marks, and clock time

A finite relative-book state need not confine the reference price to a bounded grid. Let Z_t describe spread, relative queue depths, and a possible hidden regime. At state z , event channel e has rate $\lambda_e(z)$, next state $T_e(z)$, and conditional mean price increment $\bar{j}_e(z)$. Price increments may occur even when the relative state is unchanged. With between-event price drift $b_0(z)$, define

$$(Qu)_z = \sum_e \lambda_e(z)[u_{T_e(z)} - u_z], \quad dz = b_0(z) + \sum_e \lambda_e(z)\bar{j}_e(z). \quad (17)$$

Assume finitely many states and channels, bounded rates and conditional second moments of marks, and an integrable martingale component of the price. Then compensation and the Markov property give

$$\mu_h(t) = \mathbb{E}[P_{t+h} - P_t \mid \mathcal{G}_t] = \pi_t \int_0^h e^{sQ} ds. \quad (18)$$

Indeed, the compensated price change has conditional mean zero, and $\mathbb{E}[d(Z_{t+s}) \mid \mathcal{G}_t] = \pi_t e^{sQ} d$. The vector d is the total price compensator. In the earlier bounded price-vector model, $d = Qp$, so integrating the semigroup derivative recovers (14) when $\pi_t p = p_t$.

Order-book event models of this type build on established Markov and queue-reactive constructions (Cont et al., 2010; Huang et al., 2015). The distinction between a relative state and an additive price is essential to their use in a forecast certificate. The generator of the relative state alone does not specify the price marks or the event clock.

Example 5.4 (The clock and the marks cannot be discarded). Consider a single relative state, with unit upward price events of rate λp and downward events of rate $\lambda(1-p)$. The relative-state generator is $Q = 0$, but $\mu_h = h\lambda(2p-1)$. For $p = 3/4$ and $h = 1$, rates $\lambda = 1$ and $\lambda = 4$ give means $1/2$ and 2 . The relative-state generator and the upward probability per event are identical in the two models. Neither object alone determines a clock-time price forecast.

For a vector u , write $\text{span}(u) = \max_i u_i - \min_i u_i$. A stochastic semigroup always contracts span. When a stronger bound

$$\text{span}(e^{sQ}u) \leq e^{-\omega s} \text{span}(u), \quad s \geq 0, \quad (19)$$

is available for some $\omega > 0$, it improves the horizon dependence. We allow $\omega = 0$, which requires no mixing assumption, and set

$$A_\omega(h) = \int_0^h e^{-\omega s} ds = \begin{cases} (1 - e^{-\omega h})/\omega, & \omega > 0, \\ h, & \omega = 0, \end{cases} \quad (20)$$

$$C_\omega(h) = \int_0^h A_\omega(s) ds = \begin{cases} h/\omega - (1 - e^{-\omega h})/\omega^2, & \omega > 0, \\ h^2/2, & \omega = 0. \end{cases} \quad (21)$$

Proposition 5.5 (Forecast transfer for an additive price). *Let $Q, \hat{Q}$ be valid generators, $\pi, \hat{\pi}$ probability rows, and $d, \hat{d}$ price-compensator vectors on the same finite state space. Suppose Q satisfies (19). With*

$$D_d = \frac{1}{2} \text{span}(d), \quad \delta_\pi = \|\pi - \hat{\pi}\|_1, \quad \delta_Q = \|Q - \hat{Q}\|_{\infty \rightarrow \infty}, \quad \varepsilon_d = \|d - \hat{d}\|_\infty,$$

the forecasts defined by (18) satisfy

$$|\mu_h - \hat{\mu}_h| \leq h\varepsilon_d + D_d\{A_\omega(h)\delta_\pi + C_\omega(h)\delta_Q\}. \quad (22)$$

Proof. The posterior term at time s is bounded by $D_d e^{-\omega s} \delta_\pi$, because a zero-sum row annihilates constants and can be applied to the centered vector $e^{sQ} d$. For the generator term, use

$$(e^{s\hat{Q}} - e^{sQ})d = \int_0^s e^{(s-v)\hat{Q}}(\hat{Q} - Q)e^{vQ}d \, dv.$$

The left semigroup contracts the sup norm, the rate difference annihilates constants, and (19) bounds the centered right vector by $D_d e^{-\omega v}$. Thus this term is at most $D_d \delta_Q A_\omega(s)$. The drift-vector error is at most ε_d at every time under the estimated semigroup. Integrating the three bounds proves (22). $\square$

Applying Theorem 3.1 squares this forecast bound and divides by $2a$. A bounded position domain allows the sharper envelope $C_{a,W}$ of Proposition 3.3. A positive ω is an additional verified property, not a consequence of merely having a finite generator. It changes the generator contribution from quadratic to asymptotically linear in h ; a constant compensator error still accumulates linearly.

Proposition 5.6 (Sharp horizon and mixing coefficients). *Each of the three coefficients $h, A_\omega(h), C_\omega(h)$ in (22) is individually best possible over the stated class. The generator coefficient is approached by two-state marked-event models.*

Proof. Fix $D > 0, \omega \geq 0, h > 0$, and use

$$Q_0 = \begin{pmatrix} 0 & 0 \\ \omega & -\omega \end{pmatrix}, \quad d = \begin{pmatrix} 0 \\ D \end{pmatrix}, \quad \pi = (1, 0).$$

Its span contraction is exactly $e^{-\omega s}$. Changing only the posterior to $(1 - \eta, \eta)$, with $0 \leq \eta \leq 1$, changes the forecast by $D\eta A_\omega(h)$. Since $D_d = D/2$ and $\delta_\pi = 2\eta$, this attains the posterior term. Changing only the compensator to $d + c\mathbf{1}$ changes the forecast by hc , attaining the drift term in absolute value.

Now leave d, π unchanged and set

$$Q_\epsilon = \begin{pmatrix} -\epsilon & \epsilon \\ \omega & -\omega \end{pmatrix}, \quad \epsilon > 0.$$

The baseline forecast is zero, while the perturbed forecast is

$$D \int_0^h \frac{\epsilon}{\omega + \epsilon} (1 - e^{-(\omega + \epsilon)s}) \, ds = D\epsilon C_{\omega + \epsilon}(h).$$

Here $\delta_Q = 2\epsilon$, so the ratio of actual error to the generator term $D\epsilon C_\omega(h)$ tends to one as $\epsilon \downarrow 0$. This also covers $\omega = 0$. Both models have a marked-event realization: the states describe two liquidity regimes, their transitions carry no price mark, and unit upward price events occur at rate D in the second state without changing that state. $\square$

With $\omega = 0$, the last construction has forecast error $D\epsilon h^2/2 + O(\epsilon^2)$ for fixed h . Under unconstrained quadratic friction its decision loss is $D^2\epsilon^2 h^4/(8a) + O(\epsilon^3)$. Thus the fourth power of the horizon in this nonmixing generator-to-loss bound is attainable locally; it is not solely an artifact of applying two inequalities.

5.4 Filtering hidden states from events and quiet periods

The preceding results require a posterior. We now specify a finite event-observation model and give a pathwise error allowance for that posterior. Let U be the generator of unobserved state transitions. For each observed event type e , let M_e be a nonnegative matrix: $M_e(i, j)$ is the rate of an event of type e that also changes the state from i to j . Diagonal entries are permitted, because an observed event can leave the hidden state unchanged. Define

$$\nu = \sum_e M_e \mathbf{1}, \quad K = U - \text{diag}(\nu), \quad Q = K + \sum_e M_e. \quad (23)$$

Then K is a subgenerator for evolution without an observed event, and Q is the generator of the full latent state. These point-process filtering principles are classical; see [Snyder \(1972\)](#). The formulas here retain simultaneous event and state changes, as an order-book observation can require.

Proposition 5.7 (Event-history posterior and its perturbation). *Starting with prior π_0 , suppose the observed history by time t consists of events $e_1, \dots, e_N$ at ordered times, with successive waiting intervals $\Delta_1, \dots, \Delta_{N+1}$ summing to t . The posterior is the normalized row*

$$\alpha = \pi_0 e^{K\Delta_1} M_{e_1} e^{K\Delta_2} \dots M_{e_N} e^{K\Delta_{N+1}}, \quad \pi_t = \frac{\alpha}{\alpha \mathbf{1}}, \quad (24)$$

provided its denominator is positive. This denominator is a history density with respect to event times and types, not the probability of an exact continuous-time history.

For two valid models, choose common constants $a_e > 0$ at least as large as every row sum of both M_e and $\widehat{M}_e$, and put $B_e = M_e/a_e$, $\widehat{B}_e = \widehat{M}_e/a_e$. Let L be the product in (24) with B_e in place of M_e , and let $\widehat{L}$ be its fitted counterpart. Define the rescaled likelihoods

$$Z = \pi_0 L \mathbf{1}, \quad \widehat{Z} = \widehat{\pi}_0 \widehat{L} \mathbf{1}$$

and the supplied bounds

$$\|\pi_0 - \widehat{\pi}_0\|_1 \leq \varepsilon_0, \quad \|K - \widehat{K}\|_{\infty \rightarrow \infty} \leq \varepsilon_K, \quad \|B_e - \widehat{B}_e\|_{\infty \rightarrow \infty} \leq \varepsilon_{B,e}.$$

If $Z, \widehat{Z} > 0$, then

$$\|\pi_t - \widehat{\pi}_t\|_1 \leq \varepsilon_\pi(t) := \min \left\{ 2, \frac{2}{\widehat{Z}} \left(\varepsilon_0 + t\varepsilon_K + \sum_{k=1}^N \varepsilon_{B,e_k} \right) \right\}. \quad (25)$$

Proof. Before an observed event, unnormalized conditional mass evolves by $\dot{\alpha} = \alpha K$; survival requires accounting for the state-dependent observed-event hazard. At an event of type e , its likelihood and any simultaneous state transition multiply that row by M_e . Iteration proves (24). Dividing each event matrix by a common positive scalar changes the unnormalized row by a scalar and leaves its normalization unchanged.

The matrices $B_e, e^{K\Delta}$, and their fitted versions are substochastic. They contract row ℓ^1 norms, and their induced sup norms are at most one. Duhamel's formula gives $\|e^{K\Delta} - e^{\widehat{K}\Delta}\|_{\infty \rightarrow \infty} \leq \Delta \varepsilon_K$. Telescoping the product therefore yields

$$\|\pi_0 L - \widehat{\pi}_0 \widehat{L}\|_1 \leq E_{\text{hist}} := \varepsilon_0 + t\varepsilon_K + \sum_{k=1}^N \varepsilon_{B,e_k}.$$

In particular $|Z - \widehat{Z}| \leq E_{\text{hist}}$. With $u = \pi_0 L$ and $\widehat{u} = \widehat{\pi}_0 \widehat{L}$,

$$\|u/Z - \widehat{u}/\widehat{Z}\|_1 \leq \frac{|\widehat{Z} - Z| + \|u - \widehat{u}\|_1}{\widehat{Z}} \leq \frac{2E_{\text{hist}}}{\widehat{Z}}.$$

Two probability vectors are at most two apart, giving the minimum. $\square$

The common scaling constants make the product bound explicit; they do not turn the rescaled likelihood into a probability of the exact history. Rare histories can make the bound uninformative. If $\hat{Z} = 0$, the fitted posterior is undefined, so an additional support or regularization convention is needed. No uniform filtering stability or hidden-regime identifiability is asserted without additional assumptions. Observed price changes and other marks used by the trader must be included in the observation history. Known past actions can be conditioned on in a controlled event model, but the future forecasts here use the declared exogenous law.

Example 5.8 (Silence is an observation). There are two fixed hidden regimes, fast and slow, with observed event rates 3 and 1 and equal prior probabilities. Here $U = 0$, $M = \text{diag}(3, 1)$, and $K = -\text{diag}(3, 1)$. No event for one unit of time gives

$$\mathbb{P}(\text{fast} \mid \text{no event by } 1) = \frac{e^{-3}}{e^{-3} + e^{-1}} = \frac{1}{1 + e^2} \approx 0.119203.$$

In contrast, an event observed with negligible waiting time from the equal prior gives fast-regime probability $3/4$. If each event carries a unit upward price mark, the expected price increase over the next unit of time after the quiet period is approximately 1.238406, compared with the unconditional value 2. This synthetic example isolates the information in waiting times; it is not a calibrated price model.

5.5 State reduction and a complete decision certificate

Queue features are often compressed before forecasting. Equality of displayed depth does not ensure equality of future event laws: two tagged orders can have different quantities ahead, and a hidden regime can change event rates without changing the snapshot. We quantify the effect of a specified finite state reduction.

Let $c : \{1, \dots, n\} \rightarrow \{1, \dots, m\}$ be a surjective cell map and let $L_{i\ell} = \mathbf{1}_{\{c(i)=\ell\}}$ lift a coarse vector to the full state space. In this subsection L denotes this lift, not the likelihood product in Proposition 5.7. For a valid coarse generator $\bar{Q}$, put

$$R = QL - L\bar{Q}, \quad \delta_R = \|R\|_{\infty \rightarrow \infty}. \quad (26)$$

The following intertwining criterion is the finite-chain strong lumpability criterion (Buchholz, 1994); it is included to make the state-reduction assumption checkable.

Proposition 5.9 (Exact and approximate state reduction). *The identity $e^{sQ}L = Le^{s\bar{Q}}$ for every $s \geq 0$ holds if and only if $R = 0$. Equivalently, for every full state i and coarse cell b ,*

$$\sum_{j:c(j)=b} Q_{ij} = \bar{Q}_{c(i),b}. \quad (27)$$

In general,

$$e^{sQ}L - Le^{s\bar{Q}} = \int_0^s e^{(s-v)Q} R e^{v\bar{Q}} dv. \quad (28)$$

Proof. Differentiate the exact identity at zero to obtain $R = 0$. Conversely, $QL = L\bar{Q}$ implies $Q^k L = L\bar{Q}^k$ for every integer $k \geq 0$, hence the semigroup identity by its power series. Matrix multiplication gives (27). Differentiating $e^{(s-v)Q} L e^{v\bar{Q}}$ in v and integrating proves (28). $\square$

Thus aggregate rates to every coarse cell must agree within each cell of the full state space. Even when this condition holds, a coarse price forecast also requires compatibility of the price compensator; state-transition lumpability alone omits marked events that leave the state unchanged.

Corollary 5.10 (When filtering can also be reduced). *In the event-observation model of Proposition 5.7, suppose a valid coarse observation model satisfies*

$$KL = L\bar{K}, \quad M_e L = L\bar{M}_e \quad \text{for every observed type } e.$$

Starting from coarse prior $\pi_0 L$, its event-history posterior equals $\pi_t L$ whenever the history has positive likelihood.

Proof. The first equality implies $e^{sK} L = L e^{s\bar{K}}$. Successively commuting the lift through the event-history product gives $\alpha L = \bar{\alpha}$. Since $L\mathbf{1} = \mathbf{1}$, the two likelihood normalizers agree. Normalizing therefore gives the claimed posterior identity. $\square$

The forecast reduction below does not by itself authorize running the filter on a compressed state. It uses a supplied full-state posterior, which may then be aggregated. The additional conditions in Corollary 5.10 give one exact way to reduce the filter too. Lumpability of $Q = K + \sum_e M_e$ alone does not impose the separate equalities for survival and each observed event type.

Theorem 5.11 (From event observations to position loss). *Let π_t, Q, d describe the full conditional forecast (18). Let $\hat{\pi}_t$ be a fitted posterior and $(\bar{Q}, \bar{d})$ a coarse forecast model, with deployed forecast*

$$\hat{\mu}_h = (\hat{\pi}_t L) \int_0^h e^{s\bar{Q}} \bar{d} \, ds. \quad (29)$$

Suppose the coarse semigroup satisfies the span contraction (19) with $\bar{Q}$ in place of Q . Set $\bar{D} = \text{span}(\bar{d})/2$ and assume

$$\|\pi_t - \hat{\pi}_t\|_1 \leq \varepsilon_\pi, \quad \|d - L\bar{d}\|_\infty \leq \delta_d, \quad \|QL - L\bar{Q}\|_{\infty \rightarrow \infty} \leq \delta_R.$$

Then

$$|\mu_h - \hat{\mu}_h| \leq E_h := h\delta_d + \bar{D}\{A_\omega(h)\varepsilon_\pi + C_\omega(h)\delta_R\}. \quad (30)$$

In particular, ε_π may be supplied by (25). If a fitted full model satisfies $\|Q - \hat{Q}\|_{\infty \rightarrow \infty} \leq \varepsilon_Q$ and $\|d - \hat{d}\|_\infty \leq \varepsilon_d$, valid computable choices are

$$\delta_R = \|\hat{Q}L - L\bar{Q}\|_{\infty \rightarrow \infty} + \varepsilon_Q, \quad \delta_d = \|\hat{d} - L\bar{d}\|_\infty + \varepsilon_d. \quad (31)$$

Let F be the known proper closed a -strongly convex trading friction, including its convex constraints, and assume the true optimizer w^ exists. For an implemented feasible position $\tilde{w}$, suppose a subgradient $g \in \partial F(\tilde{w})$ is available and $|g - \hat{\mu}_h| \leq \chi$. If the effective domain of F has diameter at most W , the actual objective loss satisfies*

$$[\mu_h w^* - F(w^*)] - [\mu_h \tilde{w} - F(\tilde{w})] \leq C_{a,W}(E_h + \chi). \quad (32)$$

On an unbounded domain the right side can be replaced by $(E_h + \chi)^2/(2a)$. An exact fitted optimizer has $\chi = 0$.

Proof. First replace d by $L\bar{d}$ inside the true forecast; sup-norm contraction bounds the resulting error by $h\delta_d$. Next use (28). The rows of R sum to zero, so R annihilates constants. Coarse span contraction bounds the centered vector $e^{v\bar{Q}}\bar{d}$ by $\bar{D}e^{-\omega v}$, while the full semigroup contracts the sup norm. Integrating over v , then over the forecast horizon, gives $\bar{D}C_\omega(h)\delta_R$. The remaining posterior difference has zero total mass and is bounded by $\bar{D}A_\omega(h)\varepsilon_\pi$. This proves (30). Since $\|L\|_{\infty \rightarrow \infty} = 1$, the triangle inequality gives (31).

For the final step, strong convexity at $\tilde{w}$ gives, with $v = w^* - \tilde{w}$,

$$[\mu_h w^* - F(w^*)] - [\mu_h \tilde{w} - F(\tilde{w})] \leq (\mu_h - g)v - \frac{a}{2}v^2 \leq (E_h + \chi)|v| - \frac{a}{2}|v|^2.$$

Maximizing over $0 \leq |v| \leq W$ yields $C_{a,W}(E_h + \chi)$; maximizing over $|v| \geq 0$ yields the unbounded-domain bound. At an exact fitted optimum, $\hat{\mu}_h \in \partial F(\tilde{w})$, so $\chi = 0$. $\square$

The certificate keeps event-history inference, model uncertainty, state reduction, price marks, and optimizer accuracy as distinct quantities. It is conditional on valid event and generator models and supplied parameter-error bounds. The filtering formula does not estimate those bounds from data, and a small in-sample forecast error does not certify them. If trading materially changes the future book law, that effect must be modeled in the control problem. Here the book forecast is exogenous to the chosen position, and known trading costs enter F ; Section 6 treats error in an impact operator on a specified common probability space.

Example 5.12 (An event-history certificate in trading units). Consider four hidden liquidity states, reduced to cells $\{1, 2\}$ and $\{3, 4\}$. Let

$$Q_0 = \begin{pmatrix} -1.6 & 1 & 0.4 & 0.2 \\ 0.3 & -0.9 & 0.1 & 0.5 \\ 0.2 & 0.2 & -1.1 & 0.7 \\ 0.3 & 0.1 & 0.8 & -1.2 \end{pmatrix}, \quad E_{24} = 1, \quad E_{22} = -1,$$

with all other entries of E zero. The unobserved generators are $U = Q_0 + 0.02E$ and $\hat{U} = Q_0 + 0.023E$. Observed events have unit price marks $+1$ and -1 , with

$$M_+ = \text{diag}(1.5, 1.56, 0.5, 0.52), \quad M_- = \text{diag}(0.65, 0.66, 1.2, 1.22).$$

Set $\hat{M}_+ = 1.001M_+$ and $\hat{M}_- = 0.999M_-$. Because these observed events leave the hidden state unchanged, $Q = U$ and $\hat{Q} = \hat{U}$, while $d = (M_+ - M_-)\mathbf{1}$ and $\hat{d} = (\hat{M}_+ - \hat{M}_-)\mathbf{1}$.

The true and fitted priors are $(0.4, 0.1, 0.3, 0.2)$ and $(0.401, 0.099, 0.3, 0.2)$. Observe events $+, -, +$ with waiting intervals $0.08, 0.11, 0.04$, followed by a quiet interval 0.09 . Use the smallest common row-sum upper bounds as the constants a_e in Proposition 5.7. If L is the cell lift, set $A = L^\top/2$, $\bar{Q} = A\hat{Q}L$, and $\bar{d} = A\hat{d}$. The coarse two-state contraction rate is $\omega = \bar{Q}_{12} + \bar{Q}_{21} = 1.0115$.

For horizon $h = 0.7$, curvature $a = 0.7$, starting position $w_0 = 0.04$, and friction

$$F(w) = 0.35w^2 + 0.02|w - 0.04| + I_{[-0.6, 0.6]}(w),$$

deploy the exact coarse-forecast optimizer plus a position error of 0.007 . This position is interior, and its subgradient residual is $\chi = 0.0049$. Direct matrix exponentials and convex optimization give the following quantities, rounded for display:

Quantity	Value
Rescaled fitted history likelihood $\hat{Z}$	0.15267510
Posterior error allowance ε_π	0.09439656
Rate-reduction allowance δ_R	0.02900000
Price-compensator allowance δ_d	0.02725500
True forecast μ_h	0.21587418
Deployed forecast $\hat{\mu}_h$	0.21629237
Forecast error allowance E_h	0.06085928
Actual decision loss	0.00002020
Certified decision loss $C_{a,1.2}(E_h + \chi)$	0.00308877

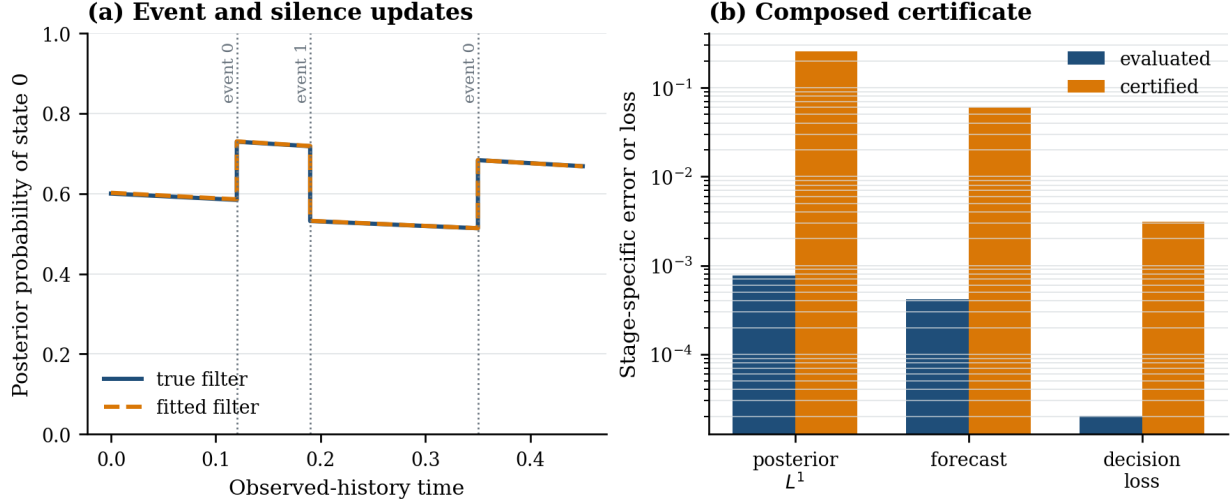


Figure 3: Event filtering and a composed decision certificate in two complementary synthetic cells. Panel (a) shows true and fitted two-state posteriors along a fixed observed history: events cause likelihood jumps, while each quiet interval evolves the posterior continuously. Panel (b) compares the independently evaluated posterior error, forecast error, and decision loss with their corresponding allowances in the verified event-to-position calculation. The stages have different units and are compared only within each pair; no market data or fitted confidence level is asserted.

The bound uses uniform parameter enclosures and a pathwise filtering allowance, so it need not be attained in this example. All rates, priors, costs, and marks in this example are synthetic, and the uncertainty allowances are computed from its two specified models. The example checks a conditional certificate, not a statistical confidence set learned from market data.

Figure 3 visualizes the two stages that are easy to conflate: history filtering and downstream loss certification.

5.6 State reduction specific to the price forecast

Preserving every transition probability is stronger than preserving the conditional mean used by a particular trading objective. Let L lift functions on a partition into vectors constant on each cell, as above. Write

$$v_h = \int_0^h e^{sQ} ds.$$

The feature preserves the horizon- h conditional mean exactly when $v_h \in \text{im } L$. This criterion concerns the specified additive price compensator and the known, common friction F . It does not assert that conditional variances, fill probabilities, or inventory-dependent rewards are preserved.

Proposition 5.13 (A finite test for price-mean sufficiency). *For a generator on n states, the following are equivalent:*

$$v_h \in \text{im } L \quad \text{for every } h \geq 0; \quad Q^j d \in \text{im } L, \quad j = 0, \dots, n-1.$$

Thus the feature need only preserve the subspace generated by $d, Qd, \dots, Q^{n-1}d$, rather than all vectors constant on its cells. Full transition lumpability implies this condition when $d \in \text{im } L$, but the converse can fail.

Proof. The map $h \mapsto v_h$ is analytic, with $v_h = \sum_{j \geq 0} h^{j+1} Q^j d / (j+1)!$. If v_h belongs to the closed linear subspace $\text{im } L$ for every $h \geq 0$, all its right derivatives at zero do too. These derivatives give every $Q^j d$. Conversely, the Cayley–Hamilton identity expresses all powers of Q of order at least n as combinations of its first n powers. The finite condition therefore places every term of the convergent series in $\text{im } L$. Under full lumpability, that image is Q -invariant, giving the stated sufficient condition. $\square$

For approximate reduction it is similarly useful to apply the intertwining residual to the price forecast before taking a norm.

Proposition 5.14 (A price-specific reduction allowance). *Let $Q, \bar{Q}$ be valid full and coarse generators, $R = QL - L\bar{Q}$, and $\bar{v}_h = \int_0^h e^{s\bar{Q}} \bar{d} \, ds$. Suppose*

$$\|d - L\bar{d}\|_\infty \leq \delta_d, \quad \|\pi - \hat{\pi}\|_1 \leq \varepsilon_\pi.$$

Define $\rho(s) = \|Re^{s\bar{Q}} \bar{d}\|_\infty$. Then

$$|\pi v_h - \hat{\pi} L \bar{v}_h| \leq h \delta_d + \int_0^h (h-s) \rho(s) \, ds + \frac{1}{2} \text{span}(\bar{v}_h) \varepsilon_\pi. \quad (33)$$

If Q is unknown but $\|Q - \hat{Q}\|_{\infty \rightarrow \infty} \leq \varepsilon_Q$, one may replace $\rho(s)$ by the computable upper bound

$$\|(\hat{Q}L - L\bar{Q})e^{s\bar{Q}} \bar{d}\|_\infty + \frac{\varepsilon_Q}{2} \text{span}(e^{s\bar{Q}} \bar{d}).$$

The resulting forecast allowance can be used in Theorem 5.11, including its optimizer residual.

Proof. Replacing d by $L\bar{d}$ costs at most $h\delta_d$. Apply the exact intertwining identity (28) to $\bar{d}$, integrate over the forecast horizon, and use sup-norm contraction of the full semigroup. Reversing the order of integration bounds the reduction term by $\int_0^h (h-s) \rho(s) \, ds$. The posterior difference has zero total mass; centering $L\bar{v}_h$ gives its half-span allowance. Finally $Q - \hat{Q}$ annihilates constants. Apply it to the centered lifted coarse vector to obtain the computable replacement. $\square$

Example 5.15 (A non-Markov feature with every mean forecast intact). Consider the four-state generator, compensator, and partition

$$Q = \begin{pmatrix} -3 & 1 & 1 & 1 \\ 1 & -5 & 2 & 2 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & 0 & -2 \end{pmatrix}, \quad d = (0, 0, 1, -1)^\top, \quad \{1, 2\}, \{3\}, \{4\}.$$

Transitions carry no price mark; unit upward price events occur in state 3 and unit downward events in state 4, each at rate one and without changing the relative state. States 1 and 2 have different transition rates to each of cells $\{3\}$ and $\{4\}$. The partition is therefore not lumpable. Nevertheless $Qd = -2d$, so

$$v_h = A_2(h)d$$

is constant on every cell for every horizon.

A valid coarse generator and compensator are

$$\bar{Q} = \begin{pmatrix} -3 & 1.5 & 1.5 \\ 2 & -2 & 0 \\ 2 & 0 & -2 \end{pmatrix}, \quad \bar{d} = (0, 1, -1)^\top.$$

Here $\|QL - L\bar{Q}\|_{\infty \rightarrow \infty} = 2$, but $(QL - L\bar{Q})e^{s\bar{Q}}\bar{d} = 0$ for every s . The price-specific reduction allowance is exactly zero. The full transition error is nonzero; it cancels in the particular conditional mean used for the position decision.

If only q_{13} is raised by 0.03, with its diagonal adjusted, the original coarse model has the following errors:

Horizon	Actual maximum mean error	Price-specific allowance
0.1	0.00012741	0.00014049
0.5	0.00180691	0.00275910
1.0	0.00420853	0.00851502

The allowance now uses the residual acting on the coarse price vector, rather than the full transition norm. Entries in its column are rounded upward for display. Exact rational algebra checks the unperturbed subspace identities; independent matrix exponentials and quadrature evaluate the perturbed comparison.

5.7 Enclosing the forecast before choosing a position

The norm certificate can be conservative because it centers vectors, takes a worst-state norm, and bounds a normalized posterior perturbation before evaluating the forecast. An alternative keeps positive likelihood bounds and encloses the future price increment directly. It uses the same finite event model; no mixing assumption is required.

Let $K^- \leq K^\theta \leq K^+$ be componentwise bounds on the no-observation generator, with $K^\pm$ Metzler matrices (nonnegative off-diagonal entries). They need not themselves be valid no-observation generators. Let $0 \leq M_e^- \leq M_e^\theta \leq M_e^+$ bound every observed-event matrix, and let $0 \leq p^- \leq p^\theta \leq p^+$ bound the prior. For the observed history with event labels $e_1, \dots, e_N$ and waiting intervals $\Delta_1, \dots, \Delta_{N+1}$, form

$$\alpha^\pm = p^\pm e^{\Delta_1 K^\pm} M_{e_1}^\pm e^{\Delta_2 K^\pm} \dots M_{e_N}^\pm e^{\Delta_{N+1} K^\pm}.$$

The prior endpoints need not sum to one; the true prior does. All rates and event matrices must still contain at least one valid model. These products are unnormalized row vectors.

For the future law, suppose every off-diagonal generator entry satisfies $q_{ij}^\theta \in [q_{ij}^-, q_{ij}^+]$, with $q_{ij}^- \geq 0$, and the compensator is bounded by $d^- \leq d^\theta \leq d^+$. Choose $\Lambda > 0$ such that $\sum_{j \neq i} q_{ij}^+ \leq \Lambda$ for every row. For any real vector v , define

$$(P^\pm v)_i = v_i + \frac{1}{\Lambda} \sum_{j \neq i} \text{ext}^\pm \{q_{ij}^-(v_j - v_i), q_{ij}^+(v_j - v_i)\}. \quad (34)$$

Here $\text{ext}^- = \min$ and $\text{ext}^+ = \max$. Set $v_0^- = d^-$, $v_0^+ = d^+$ and $v_{k+1}^\pm = P^\pm v_k^\pm$. Define positive weights

$$b_k(h) = \int_0^h e^{-\Lambda s} \frac{(\Lambda s)^k}{k!} ds = \frac{1 - e^{-\Lambda h} \sum_{j=0}^k (\Lambda h)^j / j!}{\Lambda}, \quad (35)$$

$$t_J(h) = h - \sum_{k=0}^J b_k(h), \quad D = \max_i \max\{|d_i^-|, |d_i^+|\},$$

$$v_h^- = \sum_{k=0}^J b_k(h) v_k^- - D t_J(h) \mathbf{1}, \quad v_h^+ = \sum_{k=0}^J b_k(h) v_k^+ + D t_J(h) \mathbf{1}. \quad (36)$$

Theorem 5.16 (An event-history forecast interval). *Under these componentwise enclosures,*

$$\alpha^- \leq \alpha^\theta \leq \alpha^+, \quad v_h^- \leq \int_0^h e^{sQ^\theta} d^\theta ds \leq v_h^+ \quad (\theta \in \Theta).$$

If $\sum_i \alpha_i^- > 0$, then the conditional forecast belongs to the computable interval $[l_h, u_h]$, where

$$l_h = \min_{\alpha^- \leq y \leq \alpha^+} \frac{\sum_i y_i (v_h^-)_i}{\sum_i y_i}, \quad u_h = \max_{\alpha^- \leq y \leq \alpha^+} \frac{\sum_i y_i (v_h^+)_i}{\sum_i y_i}. \quad (37)$$

Each extremum can be computed by sorting its scores and evaluating at most $m + 1$ weight vectors, where m is the number of states. For the maximum, use lower weights below a cut in the sorted scores and upper weights above it; for the minimum reverse these assignments and take the smallest ratio. Consequently (6) bounds the loss of every fixed feasible deployed position, and Theorem 3.4 gives a minimax position for this certified forecast interval.

Proof. Metzler exponentials are nonnegative and preserve componentwise matrix order: after a common diagonal shift, this follows term by term from the exponential series of nonnegative matrices. Multiplication by nonnegative matrices and row vectors preserves order. Induction through the history proves the first enclosure.

For each valid Q^θ , $P^\theta = I + Q^\theta/\Lambda$ is stochastic. The two operators in (34) are the rowwise extrema of stochastic linear maps over the rate box. They are therefore monotone, and $P^-v \leq P^\theta v \leq P^+v$. Induction gives $v_k^- \leq (P^\theta)^k d^\theta \leq v_k^+$. Uniformization and integration yield

$$\int_0^h e^{sQ^\theta} d^\theta ds = \sum_{k=0}^{\infty} b_k(h) (P^\theta)^k d^\theta.$$

The weights sum to h , and stochasticity bounds each remaining vector in sup norm by D . The omitted terms therefore lie between $\pm Dt_J(h)\mathbf{1}$, proving (36).

The conditional mean is the ratio of the history row applied to the future vector and the sum of the history row. Its weights are nonnegative and its denominator is positive. Replacing the future vector by its lower or upper bound, and then optimizing the ratio over the weight box, proves (37). For fixed scores s_i , the derivative of the ratio with respect to weight y_i has the sign of $s_i - r$, where r is the current ratio. At a maximizing vertex, every score above the attained ratio can use its upper weight and every score below it its lower weight. Scores equal to the ratio can use either endpoint without changing the ratio. Sorting therefore reduces the candidates to the displayed cuts. The minimum follows with reversed choices. The final decision claims follow from the preceding theorem. $\square$

This method normalizes only at the final forecast calculation. It avoids replacing a rare-history likelihood by a global posterior-norm allowance, although a very small or vanishing likelihood lower bound can still make the interval uninformative. The uncertainty set has also been enlarged: the boxes for history weights and future values can discard correlations induced by one common parameter. The ratio extrema are exact over those boxes, and the position is minimax over their resulting interval. Neither statement claims sharpness over the original parameter family when those enlargements are strict.

Example 5.17 (A queue depletion attains the interval decision bound). A best-ask queue consists of one lot. Its depletion time is exponential of rate λ ; depletion raises the reference price by one tick,

and no further price event occurs before the forecast horizon. The two states are live and depleted, with generator and compensator

$$Q_\lambda = \begin{pmatrix} -\lambda & \lambda \\ 0 & 0 \end{pmatrix}, \quad d_\lambda = \begin{pmatrix} \lambda \\ 0 \end{pmatrix}.$$

Starting live, the forecast is $\mu_h = 1 - e^{-\lambda h}$. For any $0 < l < u < 1$, choosing

$$\lambda \in \left[-\frac{\log(1-l)}{h}, -\frac{\log(1-u)}{h} \right]$$

realizes every mean in $[l, u]$, including both endpoints. Thus the interval minimax lower bounds in Corollary 3.5 are attained in a price-moving queue model with this interval information. For the proportional case also choose $0 < c - \epsilon < c + \epsilon < 1$.

Example 5.18 (An interval certificate for the four-state book). Retain the event history, horizon, friction, and fitted model of Example 5.12. Specify an uncertainty box as follows. Only the off-diagonal entry q_{24} is uncertain, within 0.003 of its fitted value 0.523; all other off-diagonal rates are known, and each diagonal is minus its row's exit rate. Each diagonal entry of M_+ is within 0.002 of its fitted value, and each entry of M_- within 0.0015. The first two prior coordinates are within 0.001 of 0.401, 0.099, the other two are fixed, and valid priors sum to one. Observed events leave the hidden state unchanged. These bounds include the reference model of that example.

Construct K^- using lower off-diagonal rates and diagonal entries equal to minus the upper external exit and observed-event rates; reverse the choices for K^+ . Future compensator bounds are obtained by subtracting the upper downward-event rate from the lower upward-event rate, and conversely. Use $\Lambda = 2$ and $J = 24$ for the future series. The resulting forecast interval is

$$[l_h, u_h] = [0.210697132332, 0.222819777337].$$

All displayed endpoints are rounded outward. The fixed deployed position is 0.287417666770, the position of the earlier example rounded to 12 decimal places. The comparison is

Quantity	Value
Reference-model forecast	0.21587418
Norm-certificate loss bound for this box	0.004036766177
Interval loss bound for the deployed position	0.000078678533
Interval minimax position, rounded down	0.281083506906
Loss bound at that rounded position	0.000026242594

Both certificates in the table use exactly this uncertainty box and the same deployed position. The norm certificate is recomputed from Theorem 5.11, with $\varepsilon_Q = 0.006$, $\varepsilon_d = 0.0035$, $\delta_R = 0.029$, and $\delta_d = 0.028535$, together with the history-filter bound and the actual subgradient residual. Its value differs from Example 5.12, whose allowances compared one specified pair of models. Retaining the forecast interval reduces the same-position bound by a factor exceeding 51. Choosing the interval minimax position reduces it further. These comparisons concern guaranteed losses, not observed profitability.

The enclosures use rational arithmetic, including rigorous exponential tails. For $x \geq 0$, the Taylor polynomial of degree 44 for e^{-x} is an upper bound and that of degree 45 is a lower bound, by the sign of the Lagrange remainder. For a Metzler matrix K , choose $c > 0$ so $P = I + K/c \geq 0$.

Truncate $e^{tK} = e^{-ct} \sum_{k \geq 0} (ct)^k P^k / k!$ after degree 30. If $y = ct \|P\|_\infty < 32$, its norm tail is bounded by

$$e^{-ct} \frac{y^{31}}{31!} \frac{1}{1 - y/32},$$

with e^{-ct} replaced by its rational upper enclosure. The future weights in (35) are enclosed using the same scalar bounds, with sign-aware interval multiplication. The remaining integrated Poisson tail is below 3×10^{-24} . Outward rounding is included at each reported precision. Independent matrix-exponential evaluation of the reference model lies inside both the history and future enclosures. Exact enumeration of all 16 weight-box vertices independently checks the sorted-cut ratio calculation in this four-state example.

5.8 The decision value of hidden book information

The preceding certificates compare forecasts or implementations within a declared observation class. An additional loss appears when the benchmark observes information unavailable to the trader. Let Z be the future price increment and $\mathcal{G} \subseteq \mathcal{H}$ two information sets at the decision time. Define

$$m_{\mathcal{G}} = \mathbb{E}[Z \mid \mathcal{G}], \quad m_{\mathcal{H}} = \mathbb{E}[Z \mid \mathcal{H}].$$

Here $\mathcal{H}$ can reveal the current hidden liquidity state, while $\mathcal{G}$ contains a public feature or an event history. Take one common, known, proper closed convex friction F . Assume measurable maximizing positions exist and all the following expectations are finite. Comparison of information through its decision value is classical (Blackwell, 1953); the formulas below express that value in the same loss units as the book-model certificates.

Theorem 5.19 (Observation loss and implementation loss). *The optimal expected value with information $\mathcal{A}$ is*

$$V_{\mathcal{A}} = \mathbb{E}F^*(m_{\mathcal{A}}), \quad \mathcal{A} \in \{\mathcal{G}, \mathcal{H}\}.$$

For any feasible $\mathcal{G}$ -measurable deployed position $\tilde{w}$,

$$\begin{aligned} V_{\mathcal{H}} - \mathbb{E}[Z\tilde{w} - F(\tilde{w})] &= \mathcal{I}_F(\mathcal{H} \mid \mathcal{G}) + \mathbb{E}\ell_F(m_{\mathcal{G}}, \tilde{w}), \\ \mathcal{I}_F(\mathcal{H} \mid \mathcal{G}) &:= \mathbb{E}\{F^*(m_{\mathcal{H}}) - F^*(m_{\mathcal{G}})\} \geq 0, \\ \ell_F(m, w) &:= F^*(m) - mw + F(w) \geq 0. \end{aligned} \tag{38}$$

For any maximizing coarse-information position $w_{\mathcal{G}}$, the information term also equals $\mathbb{E}\ell_F(m_{\mathcal{H}}, w_{\mathcal{G}})$. It vanishes precisely when that coarse-information position is also optimal at $m_{\mathcal{H}}$ almost surely. If F is a -strongly convex, then

$$\mathcal{I}_F(\mathcal{H} \mid \mathcal{G}) \leq \frac{\mathbb{E}(m_{\mathcal{H}} - m_{\mathcal{G}})^2}{2a}. \tag{39}$$

For $F(w) = aw^2/2$ on $\mathbb{R}$, equality holds, and the implementation term is $\mathbb{E}(a\tilde{w} - m_{\mathcal{G}})^2/(2a)$.

Proof. Conditioning the reward on the decision information gives $\mathbb{E}[m_{\mathcal{A}}w - F(w)]$; pointwise conjugacy gives $V_{\mathcal{A}}$. Add and subtract $V_{\mathcal{G}}$ to obtain (38). Since $\mathbb{E}[m_{\mathcal{H}} \mid \mathcal{G}] = m_{\mathcal{G}}$, conditional Jensen applied to F^* proves nonnegativity of the information term.

For a maximizing $w_{\mathcal{G}}$, conjugacy gives $F(w_{\mathcal{G}}) = m_{\mathcal{G}}w_{\mathcal{G}} - F^*(m_{\mathcal{G}})$. Conditional expectation cancels $(m_{\mathcal{H}} - m_{\mathcal{G}})w_{\mathcal{G}}$. The resulting identity is the claimed expected Fenchel gap. This gap is nonnegative pointwise and zero exactly at an optimal position, proving the equality condition. Under strong convexity, the gap is the conjugate Bregman loss of Theorem 3.1, bounded by the displayed squared error. Direct substitution gives the quadratic equalities. $\square$

The decomposition is additive for general convex costs, but only the quadratic specialization is an orthogonal squared-error identity. Hidden state variation can have zero decision value: with proportional costs, all refined means may remain inside the same no-trade region. Conversely, a perfect parameter model and an exact event filter can still leave a positive information term.

For a concrete finite observation experiment, let the hidden state have prior p_i , state-conditional forecast v_i , and observation kernel $B_{io} = \mathbb{P}(O = o \mid X = i)$. Assume the observation is conditionally independent of the future price increment given X . Set

$$p_o = \sum_i p_i B_{io}, \quad m_o = \frac{\sum_i p_i B_{io} v_i}{p_o} \quad (p_o > 0).$$

Taking $\mathcal{H} = \sigma(X, O)$ and $\mathcal{G} = \sigma(O)$ gives the exact formula

$$\mathcal{I}_F = \sum_i p_i F^*(v_i) - \sum_{o: p_o > 0} p_o F^*(m_o). \quad (40)$$

For quadratic friction it is $\sum_{i,o} p_i B_{io} (v_i - m_o)^2 / (2a)$. This measures the price-relevant information lost by a noisy depth feature, not the entropy of the hidden state. Proposition 5.13 supplies a finite algebraic condition for the loss to vanish at every forecast horizon when the observation is a deterministic book partition.

5.9 The exact value of a delayed book observation

Latency costs in high-frequency trading are established (Moallemi and Sağlam, 2013). Here an exogenous finite book gives an exact distinction between the information lost during a feed delay and the additional error of treating a stale state as current. This is an observation-delay calculation. Sending an order after that decision can introduce the separate execution and cancellation delays modeled in a control problem.

Write $P_t = e^{tQ}$, and fix forecast horizon h with $v = \int_0^h P_s ds$. The trader's last observed state is X_0 ; the position is chosen at time δ , when the current state is X_δ . No additional informative events are observed in between. By the Markov property the full-information mean is $v(X_\delta)$, whereas the correct delayed-information mean is $(P_\delta v)(X_0)$. Formally, take $\mathcal{G} = \sigma(X_0)$ and $\mathcal{H} = \sigma(X_0, X_\delta)$: the benchmark retains the old state and additionally observes the current one. The initial-state prior is a row p , and vector squares below are componentwise.

Proposition 5.20 (Feed delay, state jumps, and trading loss). *Under unconstrained quadratic friction $F(w) = aw^2/2$, the information loss is exactly*

$$\mathcal{I}_a(\delta) = \frac{1}{2a} p \{P_\delta(v^2) - (P_\delta v)^2\}. \quad (41)$$

Define the jump-variance operator

$$\Gamma_Q(f)_i = \sum_{j \neq i} q_{ij} (f_j - f_i)^2 = \{Q(f^2) - 2f(Qf)\}_i.$$

Then

$$\mathcal{I}_a(\delta) = \frac{1}{2a} \int_0^\delta p P_s \Gamma_Q(P_{\delta-s} v) ds, \quad (42)$$

$$\mathcal{I}_a(\delta) = \frac{\delta}{2a} p \Gamma_Q(v) + O(\delta^2) \quad (\delta \downarrow 0). \quad (43)$$

If the trader instead uses the unpropagated stale forecast $v(X_0)$, its loss against the full-information benchmark is

$$\mathcal{I}_a(\delta) + \frac{1}{2a}p(\mathbf{P}_\delta v - v)^2. \quad (44)$$

The second term is $\delta^2 p(Qv)^2/(2a) + O(\delta^3)$.

Proof. Conditional on $X_0 = i$, the variance of $v(X_\delta)$ is $(\mathbf{P}_\delta v^2)_i - (\mathbf{P}_\delta v)_i^2$. Apply Theorem 5.19 to obtain (41). For the integral identity, differentiate

$$s \mapsto \mathbf{P}_s\{(\mathbf{P}_{\delta-s}v)^2\}.$$

Its derivative is $\mathbf{P}_s\Gamma_Q(\mathbf{P}_{\delta-s}v)$. The endpoint difference is precisely the variance vector in (41). Analyticity of the finite semigroup gives (43). Finally, conditional mean orthogonality splits $v(X_\delta) - v(X_0)$ into its centered current-state innovation and the delayed mean minus the stale forecast. Squaring and taking expectations gives (44); Taylor expansion of $\mathbf{P}_\delta v$ gives the last term. $\square$

The leading information loss is linear in the delay when $p\Gamma_Q(v) > 0$, despite the quadratic objective. A state jump has probability of order δ and can change the forecast by a finite amount. Propagating the last observed state removes the additional forecast bias; it cannot reveal the unobserved jump.

Corollary 5.21 (A stationary latency frontier). *If $pQ = 0$, then for any common convex friction with finite conjugate values on the relevant means,*

$$\mathcal{I}_F(\delta) = pF^*(v) - pF^*(\mathbf{P}_\delta v)$$

is nondecreasing in δ . If Q is irreducible, its limit is $pF^(v) - F^*(pv)$. In the quadratic case,*

$$\mathcal{I}'_a(\delta) = \frac{1}{2a}p\Gamma_Q(\mathbf{P}_\delta v) \geq 0.$$

Proof. For $s \geq 0$, conditional Jensen and $p\mathbf{P}_s = p$ give

$$pF^*(\mathbf{P}_{\delta+s}v) = pF^*(\mathbf{P}_s\mathbf{P}_\delta v) \leq p\mathbf{P}_sF^*(\mathbf{P}_\delta v) = pF^*(\mathbf{P}_\delta v).$$

Irreducibility gives $\mathbf{P}_\delta v \rightarrow (pv)\mathbf{1}$; continuity on the relevant finite interval gives the limit. Differentiate the quadratic formula and use $pQ(f^2) = 0$ to obtain its derivative. $\square$

Stationarity is essential to the monotonicity assertion; a fixed nonstationary initial state can have a different latency profile. For a random delay independent of the initial state and future book law, a known timestamp gives mean $(\mathbf{P}_D v)(X_0)$. If the timestamp is hidden, the correct mean uses $\bar{\mathbf{P}}v = \mathbb{E}_D[\mathbf{P}_D v]$. For quadratic friction, revealing the timestamp adds value

$$\frac{1}{2a}p\left\{\mathbb{E}_D[(\mathbf{P}_D v)^2] - (\bar{\mathbf{P}}v)^2\right\} \geq 0. \quad (45)$$

This follows from the same conditional-variance decomposition. In general $\bar{\mathbf{P}} \neq \mathbf{P}_{\mathbb{E}D}$, so propagating for the mean delay is a further model approximation.

Example 5.22 (A latency threshold for trading on book imbalance). Let two liquidity regimes switch symmetrically at rate $\kappa > 0$, with price compensators $+b, -b$. These compensators can be generated by state-preserving upward and downward price events. Under the stationary prior $(1/2, 1/2)$, define

$$v_h^0 = bA_{2\kappa}(h), \quad \eta_\delta = e^{-2\kappa\delta}.$$

The current-state forecast takes values $\pm v_h^0$, and the correct stale-state forecast takes values $\pm \eta_\delta v_h^0$. Thus

$$\begin{aligned} \mathcal{I}_a(\delta) &= \frac{(v_h^0)^2}{2a}(1 - e^{-4\kappa\delta}), \\ \text{additional loss from no propagation} &= \frac{(v_h^0)^2}{2a}(1 - e^{-2\kappa\delta})^2. \end{aligned}$$

For mixed friction $F(w) = aw^2/2 + c|w|$, the information loss is

$$\mathcal{I}_F(\delta) = \frac{(v_h^0 - c)_+^2 - (\eta_\delta v_h^0 - c)_+^2}{2a}.$$

If $v_h^0 > c > 0$, the optimal delayed-information position enters the no-trade region at

$$\delta_{\text{stop}} = \frac{1}{2\kappa} \log \frac{v_h^0}{c}.$$

Above this delay, an exact model optimally takes no position based on the stale regime alone.

Choose $\kappa = 20$, $b = 2$, $h = 0.05$, $a = 0.7$, and $c = 0.012$, with time measured in seconds and price marks in model units. For example, upward/downward event rates 3, 1 in one regime and 1, 3 in the other give the compensators. Then $v_h^0 = 0.04323324$ and $\delta_{\text{stop}} = 0.03204257$ seconds. The values are

Delay (ms)	Quadratic information loss	Mixed-cost information loss	Absolute mixed-cost position
0	0	0	0.04461891
1	0.00010265	0.00007359	0.04219719
5	0.00044015	0.00030580	0.03342340
10	0.00073519	0.00049085	0.02425729
20	0.00106553	0.00065741	0.01060849
40	0.00128066	0.00069680	0

The two loss columns use different objectives and should not be read as one policy comparison. Within each objective, the benchmark observes the current regime. The example is synthetic; the threshold is implied by these parameters and does not estimate a venue's profitable latency budget.

The supplement independently checks the conditional-variance integral in 36 nonreversible finite-state cases and the general observation/implementation decomposition in 48 finite experiments. The largest discrepancies are below 6×10^{-16} and 2×10^{-16} , respectively. It also checks the timestamp decomposition and the two-state closed forms against transition-matrix calculations.

Figure 4 displays the complete delay frontier for the parameters in Example 5.22.

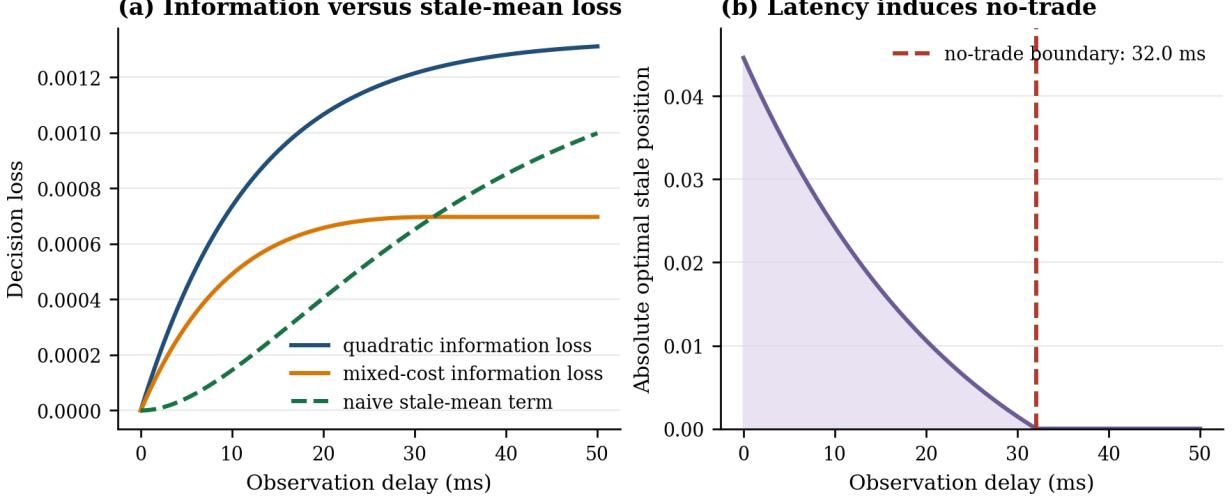


Figure 4: Exact two-state latency frontier. Panel (a) separates quadratic information loss, mixed-cost information loss, and the additional quadratic loss from using an unpropagated stale mean. The objectives differ, so the blue and orange curves are not one policy comparison. Panel (b) shows the mixed-cost position shrinking to zero at the analytic boundary $\delta_{\text{stop}} = 32.0$ ms. The calculation uses the declared synthetic parameters $\kappa = 20$, $b = 2$, $h = 0.05$, $a = 0.7$, and $c = 0.012$.

6 A joint certificate for forecasts, impact, and constraints

6.1 The adapted space and its causal adjoint

Fix a finite horizon and an exogenous probability space with filtration $(\mathcal{G}_t)_{0 \leq t \leq T}$ satisfying the usual conditions. Let $\mathbf{H} = L^2_{\text{prog}}(\Omega \times [0, T]; \mathbb{R}^d)$, modulo equality almost everywhere, with $\langle u, v \rangle = \mathbb{E} \int_0^T u_t^\top v_t dt$. It is a closed Hilbert subspace of the ambient product L^2 space. A trading rate v , from a deterministic initial position w_0 , produces position $w_t = w_0 + (Bv)_t$, where

$$(Bv)_t = \int_0^t v_s ds.$$

The operator is bounded, for example $\|B\| \leq T$. Its adjoint on this information class is

$$(B^*z)_s = \mathbb{E} \left[\int_s^T z_t dt \mid \mathcal{G}_s \right], \quad (46)$$

using a progressively measurable version. Indeed, Fubini and conditional expectation give $\mathbb{E} \int z_t^\top \int_0^t v_s ds dt = \mathbb{E} \int v_s^\top \mathbb{E}[\int_s^T z_t dt \mid \mathcal{G}_s] ds$. The required integrability follows from Cauchy–Schwarz. The conditional expectation is essential: an unconditioned future integral is the adjoint in a different, noncausal space.

Let $\mathcal{K}$ be a bounded self-adjoint positive semidefinite impact operator on $\mathbf{H}$, let $\gamma \geq 0, \eta > 0$, and let $\alpha \in \mathbf{H}$ be an exogenous expected-return-rate process. The negative expected gain, up to terms independent of v , is

$$\begin{aligned} \mathcal{J}(v) &= \frac{1}{2} \langle v, Av \rangle - \langle f, v \rangle + \Psi(v), \\ A &= 2\eta I + \mathcal{K} + \gamma B^*B, \quad f = B^*(\alpha - \gamma w_0). \end{aligned} \quad (47)$$

Here Ψ can include a proportional turnover charge $c \mathbb{E} \int \|v_t\|_1 dt$ and the indicator of a common nonempty closed convex feasible set. Pointwise rate bounds and the pathwise mandate $\int_0^T v_t dt = X$ are such constraints whenever jointly feasible. The terminal integral is a bounded linear map into $L^2(\Omega)$, so its equality constraint is closed. Every statement concerns rates; atomic endpoint blocks are outside $\mathbf{H}$.

For a scalar symmetric positive semidefinite kernel $G(|t - s|)$, with $G \in L^1(0, T)$, its adapted energy operator is

$$(\mathcal{K}v)_t = \int_0^t G(t - s)v_s ds + \mathbb{E} \left[\int_t^T G(s - t)v_s ds \mid \mathcal{G}_t \right].$$

It is the orthogonal compression of the symmetric kernel operator to $\mathbf{H}$, and is therefore self-adjoint and positive semidefinite. Its quadratic form is the expected symmetric transient-impact energy. These hypotheses give $A \succeq 2\eta I$; no coercivity assertion about the transient kernel alone is needed.

6.2 The joint residual and its exact affine form

The next result uses the abstract form (47): $A, \hat{A}$ are bounded self-adjoint operators on a common real Hilbert space, bounded below by positive constants $m, \hat{m}$; $f, \hat{f}$ belong to that space; and Ψ is a common proper lower semicontinuous convex function. The estimated objective replaces A, f by $\hat{A}, \hat{f}$. Both objectives have unique minimizers: an affine lower bound for Ψ and positive quadratic curvature give coercivity, while convex weak lower semicontinuity gives attainment.

Theorem 6.1 (Joint model-and-optimization residual). *Let v^* minimize the true objective. For a feasible implemented $\hat{v}$, suppose $\hat{\xi} \in \partial\Psi(\hat{v})$ is supplied, and define*

$$\hat{s} = \hat{A}\hat{v} - \hat{f} + \hat{\xi}, \quad r = (A - \hat{A})\hat{v} - (f - \hat{f}), \quad s = r + \hat{s}.$$

Then

$$0 \leq \mathcal{J}(\hat{v}) - \mathcal{J}(v^*) \leq \frac{\|s\|^2}{2m}, \quad \|\hat{v} - v^*\| \leq \frac{\|s\|}{m}. \quad (48)$$

In particular, if $\|A - \hat{A}\| \leq \epsilon_A$, $\|f - \hat{f}\| \leq \epsilon_f$, $\|\hat{v}\| \leq R$, and $\|\hat{s}\| \leq \tau$, then the loss is at most

$$\frac{(\epsilon_A R + \epsilon_f + \tau)^2}{2m}.$$

An exact estimated optimizer admits $\hat{s} = 0$.

Proof. The sum rule for a continuous differentiable quadratic and a proper closed convex function gives $s = A\hat{v} - f + \hat{\xi} \in \partial\mathcal{J}(\hat{v})$. Strong convexity gives, for every u ,

$$\mathcal{J}(u) \geq \mathcal{J}(\hat{v}) + \langle s, u - \hat{v} \rangle + \frac{m}{2} \|u - \hat{v}\|^2.$$

Set $u = v^*$ and complete the square to obtain the loss bound. Strong monotonicity between s and $0 \in \partial\mathcal{J}(v^*)$ gives $m\|\hat{v} - v^*\|^2 \leq \langle s, \hat{v} - v^* \rangle$, proving the distance bound. The triangle inequality proves the error-budget form. Fermat's rule for the estimated objective supplies the zero residual at its minimizer. $\square$

The estimate includes interactions between forecast and impact errors: their residuals can reinforce or cancel. Adding separate forecast and impact losses as if they were independent would generally discard that interaction. For arbitrary convex constraints the inequality is the appropriate statement; the exact quadratic equality requires an affine class.

Corollary 6.2 (Exact affine certificate and decision equivalence). *Let Ψ be the indicator of a nonempty closed affine space $\mathcal{M} = v_0 + V$, and let $\hat{v}$ be its exact estimated optimizer. With orthogonal projection P_V , put*

$$r_V = P_V[(A - \hat{A})\hat{v} - (f - \hat{f})], \quad A_V = P_V A|_V.$$

Then

$$\hat{v} - v^* = A_V^{-1} r_V, \quad \mathcal{J}(\hat{v}) - \mathcal{J}(v^*) = \frac{1}{2} \langle r_V, A_V^{-1} r_V \rangle. \quad (49)$$

The two models produce the same implemented optimum if and only if $r_V = 0$. Thus full identification of either primitive is not necessary for decision equivalence on the declared affine class.

Proof. Stationarity gives $P_V(Av^* - f) = P_V(\hat{A}\hat{v} - \hat{f}) = 0$. Subtract to obtain $A_V(\hat{v} - v^*) = r_V$. The compressed operator is self-adjoint and bounded below by m , so it has a bounded inverse. Expansion around v^* kills the linear term on V , giving $\mathcal{J}(\hat{v}) - \mathcal{J}(v^*) = \langle \hat{v} - v^*, A(\hat{v} - v^*) \rangle / 2$. Substitution proves the identity and the equivalence. $\square$

For deterministic rates, $f = \hat{f} = 0$, and fixed total mass, the projection is onto zero-mass variations and the formula is the execution-kernel residual. Retaining the signal gives the joint forecast-and-execution version. On a finite common feature class, the operators become matrices, and the subgradient residual is an ordinary constrained optimization certificate. Coefficient-space curvature also depends on the feature Gram matrix: redundant features must be removed or quotiented before an invertible compressed matrix can be claimed. These conclusions require a common probability law for the exogenous drivers and a common feasible class. A misspecified law of an endogenous controlled book is not silently absorbed by changing an impact matrix in this theorem.

6.3 The forecast error seen through feasible trajectories

Suppose only the forecast is misspecified: the operators, initial position, and affine feasible class are common. Write $z = \alpha - \hat{\alpha} \in \mathbf{H}$. The model residual then contains only the causal adjoint of z .

Corollary 6.3 (Causally filtered forecast error). *In Corollary 6.2, assume $A = \hat{A}$ and $f - \hat{f} = B^* z$. Define*

$$e = B^* z, \quad e_s = \mathbb{E} \left[\int_s^T (\alpha_t - \hat{\alpha}_t) dt \mid \mathcal{G}_s \right].$$

Then

$$\begin{aligned} \mathcal{J}(\hat{v}) - \mathcal{J}(v^*) &= \frac{1}{2} \langle P_V e, A_V^{-1} P_V e \rangle \\ &\leq \frac{\|P_V e\|^2}{2m} \leq \frac{2T^2}{m\pi^2} \|\alpha - \hat{\alpha}\|^2. \end{aligned} \quad (50)$$

The two forecasts induce the same rate, and hence the same position trajectory from the common initial position, if and only if $P_V e = 0$. The final constant is sharp over the class with $A \succeq mI$.

Proof. The projected residual is $r_V = -P_V e$, so (49) gives the identity. The inverse bound $A_V^{-1} \preceq m^{-1}I$ gives the first inequality. For the second, the integration operator on $\mathbf{H}$ has

$$\|B\| = \|B^*\| = 2T/\pi. \quad (51)$$

To see the upper bound, for each path put $w = Bv$, so $w(0) = 0$. Reflect w evenly about T onto $[0, 2T]$; the reflected function is zero at both endpoints. The Dirichlet Poincaré inequality on that

interval, obtained from its sine expansion, gives $\int_0^T |w|^2 \leq (2T/\pi)^2 \int_0^T |v|^2$. Average over paths. Equality is attained by the deterministic rate $v_t = u \cos(\pi t/(2T))$, with a fixed unit vector u . This rate is adapted to every filtration, so compression to $\mathbf{H}$ does not reduce the operator norm. Now use $\|P_V B^* z\| \leq \|B\| \|z\|$. When $V = \mathbf{H}$, $A = mI$, and $z_t = u \sin(\pi t/(2T))$, every inequality in (50) is an equality. This also proves sharpness of its constant. Decision equivalence follows from (49) and injectivity of rate integration. $\square$

The error is measured after conditional remaining-horizon integration and projection onto feasible directions. For a deterministic persistent bias $z_t = b$, $e_s = b(T - s)$ and

$$\frac{\|e\|^2}{\|z\|^2} = T^2/3 \quad (b \neq 0), \quad \sup_{z \neq 0} \frac{\|B^* z\|^2}{\|z\|^2} = 4T^2/\pi^2.$$

Thus constant bias is not a worst-case error at fixed L^2 norm. The maximizing error shape in the isotropic unconstrained example is the sine profile in the proof. With other impact operators or feasible directions, the worst shape is determined by $A_V^{-1/2} P_V B^*$, rather than by persistence alone. For the trading objective (47), taking $m = 2\eta$ gives the bound $T^2 \|\alpha - \hat{\alpha}\|^2 / (\eta \pi^2)$.

Projection can suppress error components on a restricted feasible class, precisely when $P_V B^* z = 0$. In the full adapted space $V = \mathbf{H}$, there is no nonzero adapted z with $B^* z = 0$. Indeed, write $B^* z = M - \int_0^T z_t dt$, where $M_s = \mathbb{E}[\int_0^T z_t dt \mid \mathcal{G}_s]$. If this process vanishes almost everywhere, its càdlàg version vanishes on $[0, T)$, and at T the same identity is zero by definition. The martingale M is then a continuous finite-variation process, hence constant, forcing $z = 0$ almost everywhere.

6.4 Information loss and model loss are orthogonal on affine classes

Let $\mathcal{M}_G \subset \mathcal{M}_F$ be nonempty closed affine subspaces of a common Hilbert space, representing two information classes with the same underlying quadratic objective $\mathcal{J}_0(v) = \langle v, Av \rangle / 2 - \langle f, v \rangle$, $A \succeq mI$. Let v_G^*, v_F^* be their true optimizers. Define $\|u\|_A^2 = \langle u, Au \rangle$.

Theorem 6.4 (Information projection and residual decomposition). *For every $\hat{v} \in \mathcal{M}_G$,*

$$\mathcal{J}_0(\hat{v}) - \mathcal{J}_0(v_F^*) = \frac{1}{2} \|v_G^* - v_F^*\|_A^2 + \frac{1}{2} \|\hat{v} - v_G^*\|^2. \quad (52)$$

If $\hat{v}$ is the exact estimated optimizer on $\mathcal{M}_G$, the second term equals $\langle r_G, A_G^{-1} r_G \rangle / 2$, with the projection and compression defined as in Corollary 6.2 for its direction space.

Proof. Let $V_G \subset V_F$ be the direction spaces. Stationarity of both optima implies $\langle A(v_G^* - v_F^*), z \rangle = 0$ for every $z \in V_G$. Take $z = \hat{v} - v_G^*$. Expansion around v_F^* first gives $\mathcal{J}_0(\hat{v}) - \mathcal{J}_0(v_F^*) = \|\hat{v} - v_F^*\|_A^2 / 2$. The vanished cross term splits this square into the two displayed energies. The residual form follows from Corollary 6.2. $\square$

The first term is the cost of the restricted information class for this objective, and remains with a perfectly estimated model. The second is the cost of the implemented rule within that class. This is a Hilbert projection statement, not a mutual-information formula. With general convex constraints, value differences still separate algebraically, but these orthogonal energy equalities need not hold. Both benchmarks must evaluate the same objective; enlarging a price filtration can change an economic return model, which must then be specified separately.

Example 6.5 (A two-date information and model decomposition). Let $Z = \pm 1$ with equal probability. At two dates choose rates (v_0, v_1) with $v_0 + v_1 = 1$ in every scenario, and minimize

$$\mathcal{J}_0(v) = \mathbb{E}[\tfrac{1}{2}(v_0^2 + v_1^2) - Zv_0].$$

The richer class sees Z before the first rate. Its optimum is $(1, 0)$ when $Z = 1$, and $(0, 1)$ when $Z = -1$, with value zero. In the restricted class, v_0 is chosen before Z is revealed; completion then fixes v_1 . Its optimum is the constant $(1/2, 1/2)$, with value $1/4$. Replacing the first-date signal Z by $Z + 1/5$ in the restricted optimization gives the implemented rates $(3/5, 2/5)$. Their true cost is $13/50$, and

$$\underbrace{13/50}_{\text{total loss}} = \underbrace{1/4}_{\text{information loss}} + \underbrace{1/100}_{\text{model loss}}.$$

Both energy terms are exact rational quantities. Better estimation within the restricted class can remove the second term, but not the first.

7 Causal trading of a mean-reverting target

7.1 Admissible tracking and an exact verification identity

Let x_t be a stationary OU target on the real time line:

$$dx_t = -\rho x_t dt + \sqrt{2\rho v} dB_t, \quad \rho, v > 0.$$

Its filtration is the completed natural filtration of its observed history. A trading position w is adapted, absolutely continuous with $\dot{w}_t = u_t$, and jointly stationary with x and u . Admissibility requires $\mathbb{E}(w_t^2 + u_t^2) < \infty$. For $\eta, \kappa > 0$, minimize

$$J(w) = \mathbb{E}[\eta u_t^2 + \eta \kappa^2 (w_t - x_t)^2].$$

The earlier risk notation is recovered from $\kappa^2 = \gamma \sigma^2 / (2\eta)$. The expectation is a stationary average cost per unit time.

Theorem 7.1 (Causal optimum and exact excess cost). *Set $\theta = \kappa / (\kappa + \rho)$. The optimal stationary adapted policy is*

$$u_t = \kappa(\theta x_t - w_t), \quad w_t = \kappa \theta \int_0^\infty e^{-\kappa s} x_{t-s} ds. \quad (53)$$

Its minimum cost is

$$J^* = \eta \kappa^2 v (1 - \theta^2). \quad (54)$$

Every admissible stationary rule satisfies the exact identity

$$J(w) - J^* = \eta \mathbb{E}[(u_t + \kappa(w_t - \theta x_t))^2]. \quad (55)$$

The optimal position is unique among admissible stationary solutions, up to equality almost surely.

Proof. Define the quadratic relative value function

$$\mathcal{V}(w, x) = \eta \kappa w^2 - 2\eta \kappa \theta w x + \frac{\eta \kappa^2}{2\rho} (1 - \theta^2) x^2.$$

For a control u , its generator is $\mathcal{L}^u \mathcal{V} = u \mathcal{V}_w - \rho x \mathcal{V}_x + \rho v \mathcal{V}_{xx}$. Direct expansion, using $\rho \theta = \kappa(1 - \theta)$, gives

$$\eta u^2 + \eta \kappa^2 (w - x)^2 + \mathcal{L}^u \mathcal{V} = \eta \kappa^2 v (1 - \theta^2) + \eta [u + \kappa(w - \theta x)]^2. \quad (56)$$

For an admissible stationary control, Ito's formula on a finite time interval and stationarity give $\mathbb{E} \mathcal{L}^u \mathcal{V} = 0$. The stochastic integral is square integrable because its integrand is linear in w, x , which

have finite second moments. All drift terms are integrable by Cauchy–Schwarz. Averaging (56) proves (55).

The stable causal convolution in (53) is adapted, has finite second moments and finite trading-rate cost, and makes the square zero. It therefore attains (54). Equality for any other optimizer requires the same linear differential equation. For two solutions, their difference at t equals $e^{-\kappa(t-s)}$ times its value at any earlier s . Both solutions have uniformly bounded second moments on the real time line. Letting $s \rightarrow -\infty$ forces the difference at t to be zero in L^2 . Thus the stationary optimizer is unique. $\square$

The aim is θx_t , whereas the position is the lagged convolution in (53). Indeed

$$\kappa \int_0^\infty e^{-\kappa s} \mathbb{E}_t[x_{t+s}] ds = \theta x_t.$$

This conditional future average defines the aim, not the position. The policy is the scalar continuous-time analogue of the aim-portfolio structure in Gârleanu and Pedersen (2013).

Corollary 7.2 (Using a wrong tracking rule). *Consider any stable linear rule $\dot{w}_t = -\alpha w_t + \beta x_t$, with $\alpha > 0$. Then*

$$J(w) - J^* = \eta \mathbb{E} \left[((\kappa - \alpha)w_t + (\beta - \kappa\theta)x_t)^2 \right],$$

where

$$\mathbb{E}[w_t x_t] = \frac{\beta v}{\alpha + \rho}, \quad \mathbb{E}[w_t^2] = \frac{\beta^2 v}{\alpha(\alpha + \rho)}.$$

Small errors in α, β around $(\kappa, \kappa\theta)$ cause second-order excess cost on parameter sets bounded away from instability.

Proof. Substitute the rule in (55). The stationary moment equations for wx and w^2 give the two covariances. These moments remain bounded on compact stable parameter sets, so the quadratic residual has the stated rate. $\square$

Corollary 7.3 (Misspecified persistence and friction). *Keep the true stationary target law, $\eta, \kappa, \rho, v > 0$, and $\theta = \kappa/(\kappa + \rho)$.*

1. *If only target persistence is misspecified, let $\hat{\rho} > 0$, $\hat{\theta} = \kappa/(\kappa + \hat{\rho})$, and use $\dot{w}_t = \kappa(\hat{\theta}x_t - w_t)$. Then*

$$J(w) - J^* = \eta \kappa^2 v (\hat{\theta} - \theta)^2, \quad \frac{J(w) - J^*}{J^*} = \frac{(\hat{\theta} - \theta)^2}{1 - \theta^2}. \quad (57)$$

2. *If the target's decay rate is correct but the trading rate is $\hat{\kappa} > 0$, let $\hat{\theta} = \hat{\kappa}/(\hat{\kappa} + \rho)$ and use $\dot{w}_t = \hat{\kappa}(\hat{\theta}x_t - w_t)$. Then*

$$J(w) - J^* = \eta v [(\kappa - \hat{\kappa})^2 \hat{\theta}^3 + (\hat{\kappa}\hat{\theta} - \kappa\theta)^2 + 2(\kappa - \hat{\kappa})(\hat{\kappa}\hat{\theta} - \kappa\theta)\hat{\theta}^2]. \quad (58)$$

In the fixed-risk specification $\kappa^2 = \gamma\sigma^2/(2\eta)$, an estimated friction $\hat{\eta} > 0$ corresponds to $\hat{\kappa} = \sqrt{\gamma\sigma^2/(2\hat{\eta})}$. All losses above are evaluated using the true objective coefficients.

Proof. Apply Corollary 7.2. In the first case, $(\alpha, \beta) = (\kappa, \kappa\hat{\theta})$, so the coefficient of w in the square vanishes and $\mathbb{E}x_t^2 = v$. In the second case, $(\alpha, \beta) = (\hat{\kappa}, \hat{\kappa}\hat{\theta})$, and the stationary moments are $\mathbb{E}[w_t x_t] = v\hat{\theta}^2$ and $\mathbb{E}[w_t^2] = v\hat{\theta}^3$. Expanding the residual square gives (58). The nonnegativity of this expression follows from that square, even when its cross term is negative. $\square$

For equal errors of magnitude d , with $0 < d < \rho$, the ratio of the excess cost from $\hat{\rho} = \rho - d$ to that from $\hat{\rho} = \rho + d$ is exactly

$$\left(\frac{\kappa + \rho + d}{\kappa + \rho - d}\right)^2 > 1.$$

Underestimating mean reversion is more costly in this comparison. Locally, with $d = \hat{\rho} - \rho$,

$$J(w) - J^* = \eta v \theta^4 d^2 + O(|d|^3), \quad \frac{d\theta}{d\rho} = -\theta^2/\kappa.$$

For random estimates independent of the subsequently evaluated stationary target, supported in a fixed compact positive parameter interval, this gives

$$\mathbb{E}[J(w) - J^*] = \eta v \theta^4 \mathbb{E}(\hat{\rho} - \rho)^2 + O(\mathbb{E}|\hat{\rho} - \rho|^3).$$

The leading term uses estimation MSE, including squared bias; variance alone is sufficient only for an unbiased estimate. These formulas evaluate fixed stationary policies, or an independent mixture of them, and do not assert the same result for a continuously updated estimate coupled to the live target.

7.2 Covariance, correlation, and the cost of causality

Proposition 7.4 (Three distinct transfer quantities). *For the optimal causal rule,*

$$\frac{\text{Cov}(w_t, x_t)}{v} = \theta^2, \quad \frac{\text{Var}(w_t)}{v} = \theta^3, \quad \text{Corr}(w_t, x_t) = \sqrt{\theta}.$$

If future target values are also available, the minimum tracking cost is $J_s = \eta \kappa^2 v (1 - \theta)$. Its two-sided smoothing filter has transfer function

$$H_s(\omega) = \frac{\kappa^2}{\kappa^2 + \omega^2}$$

and normalized covariance $\text{Cov}(w_t^s, x_t)/v = \theta$. Consequently

$$J^* - J_s = \eta \kappa^2 v \theta (1 - \theta), \quad \frac{J^* - J_s}{J^*} = \frac{\theta}{1 + \theta}. \quad (59)$$

Proof. For the causal policy, stationarity gives

$$0 = -(\kappa + \rho)\mathbb{E}[wx] + \kappa\theta v, \quad 0 = -2\kappa\mathbb{E}[w^2] + 2\kappa\theta\mathbb{E}[wx].$$

Thus $\mathbb{E}[wx] = v\theta^2$ and $\mathbb{E}[w^2] = v\theta^3$; division by the standard deviations gives the correlation.

For the unconstrained smoother, the spectral density of the target is $S_x(\omega) = 2\rho v/(\rho^2 + \omega^2)$. Completing the square frequency by frequency in the tracking criterion gives H_s . Equivalently, convolution with $\kappa e^{-\kappa|s|}/2$ solves the unconstrained Euler equation; stationarity and integration by parts give, for every finite-cost jointly stationary competitor,

$$J(w) - J(w^s) = \eta \mathbb{E}[(u_t - u_t^s)^2 + \kappa^2(w_t - w_t^s)^2].$$

The cross term vanishes because $\dot{u}_t^s = \kappa^2(w_t^s - x_t)$. This verifies the minimizer beyond the class of linear filters. Its impulse response has negative-time support, so it is not implementable with the observed-history filtration. The integral

$$\int_{-\infty}^{\infty} \frac{d\omega}{2\pi(\kappa^2 + \omega^2)(\rho^2 + \omega^2)} = \frac{1}{2\kappa\rho(\kappa + \rho)}$$

gives $\mathbb{E}[w^s x] = v\theta$. At the minimizing frequency response the cost integrand is $\eta \kappa^2 (1 - H_s) S_x$, hence $J_s = \eta \kappa^2 v (1 - \theta)$. Subtracting from (54) proves (59). $\square$

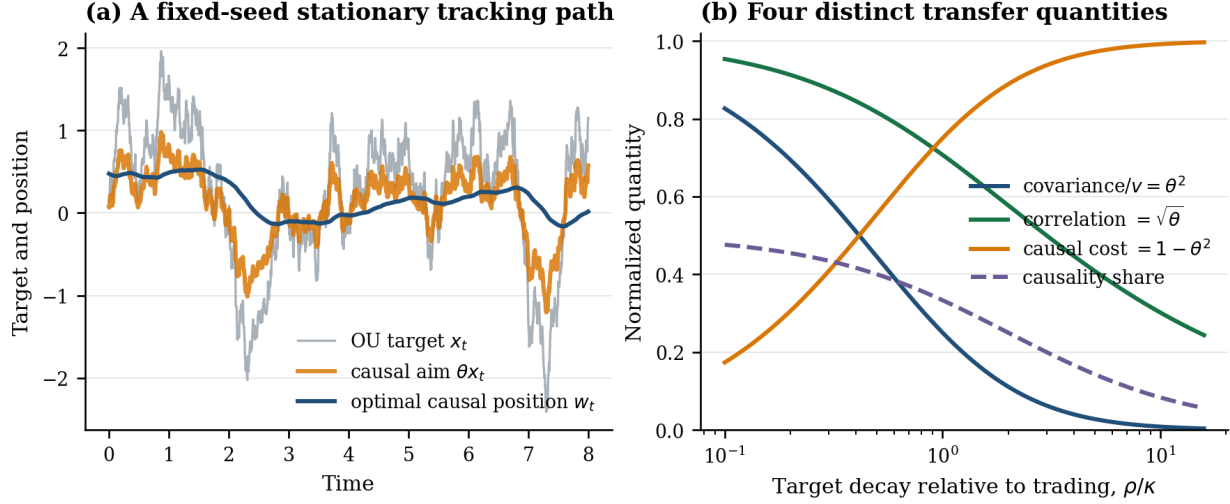


Figure 5: Causal tracking of a stationary mean-reverting target. Panel (a) shows a fixed-seed discretization with $\rho = \kappa = v = 1$, after an eight-unit burn-in: the aim is θx_t , whereas the implementable position is its lagged causal filter. Panel (b) gives the exact normalized covariance θ^2 , correlation $\sqrt{\theta}$, causal cost $1 - \theta^2$, and causality share $\theta/(1 + \theta)$ as functions of ρ/κ , where $\theta = (1 + \rho/\kappa)^{-1}$. The sample path is illustrative; the four transfer curves are analytic.

ρ/κ	Causal covariance/ v	Correlation	$J^*/(\eta\kappa^2v)$	Causality share of J^*
1	1/4	$1/\sqrt{2}$	3/4	1/3
2	1/9	$1/\sqrt{3}$	8/9	1/4
10	1/121	$1/\sqrt{11}$	120/121	1/12

The normalization matters. A target decaying twice as fast as the trading rate produces normalized covariance 1/9, Pearson correlation $1/\sqrt{3}$, and a causality increment equal to one quarter of the optimal causal tracking cost. None is a direct fraction of a strategy's profit or Sharpe ratio. Halving η at fixed $\gamma\sigma^2$ increases κ by $\sqrt{2}$; other comparative statics must account for the resulting changes in both scale and normalization.

Figure 5 contrasts an illustrative causal path with the exact transfer quantities of Proposition 7.4.

8 Information and logarithmic growth

An information bound requires an explicit portfolio opportunity set. It does not automatically cap a quadratic utility measured in different units.

Theorem 8.1 (Portfolio information ceiling). *Let the gross-return vector $R \in [\underline{r}, \bar{r}]^d$, with $0 < \underline{r} \leq \bar{r} < \infty$, and let admissible portfolios be the simplex $\Delta_d = \{b \geq 0 : \sum_i b_i = 1\}$. For a signal Y on a standard Borel space, the improvement in optimal one-period expected logarithmic growth satisfies*

$$0 \leq \mathbb{E}_Y \max_{b \in \Delta_d} \mathbb{E}[\log(b^\top R) \mid Y] - \max_{b \in \Delta_d} \mathbb{E} \log(b^\top R) \leq I(R; Y).$$

The market and feasible portfolio set are the same with and without the signal.

Proof. Let b_0 maximize the unconditional objective; compactness and the bounded positive returns ensure existence. Its directional optimality condition toward any $b \in \Delta_d$ gives

$$\mathbb{E} \frac{b^\top R}{b_0^\top R} \leq 1.$$

For a conditional law P_y and unconditional law P , the variational relative-entropy inequality implies

$$\mathbb{E}_{P_y} \log \frac{b^\top R}{b_0^\top R} \leq \text{KL}(P_y \| P) + \log \mathbb{E}_P \frac{b^\top R}{b_0^\top R} \leq \text{KL}(P_y \| P).$$

If the relative entropy is infinite the assertion is immediate. Apply the inequality to a conditional maximizing portfolio and average over Y . The expectation of the baseline conditional log return is its unconditional value. Ignoring Y gives the lower bound. $\square$

This is the classical log-optimal information comparison (Kelly, 1956; Cover and Thomas, 2006), under convenient bounded-return assumptions. In the separate finite horse-race model, with positive outcome probabilities and outcome-specific payoffs, the optimal conditional betting proportions give equality with mutual information. The simplex and positivity assumptions cannot be dropped while retaining an unrestricted assertion about arbitrary actions and arbitrary logarithmic rewards.

9 Scope and implications

The loss metric is determined jointly by the objective and the feasible set. A quadratic objective gives an exact weighted MSE. Positive curvature preserves a quadratic upper bound even with proportional costs and convex constraints. Without curvature, a threshold decision admits a uniform linear-in-error bound. Margin assumptions give sharper average rates, but a global MSE budget permits errors to concentrate at decision boundaries. The fixed-law lower construction shows that this distinction changes the best uniform exponent. When diameter and curvature coexist, their sharp combined envelope has a quadratic branch and a linear branch; the actual loss can be strictly smaller. For scalar interval uncertainty, the conjugate-secant position minimizes worst-case loss exactly without requiring curvature. The interval radius is quadratic for quadratic friction, linear at a proportional-cost threshold, and zero within a common no-trade region. Queue depletion realizes the endpoint means, so these minimax lower bounds also hold in a marked book model.

The adapted-space certificate combines forecast and impact errors before squaring the residual, and keeps the common convex constraints explicit. Its affine form identifies exactly the model discrepancies that leave the optimal decision unchanged. With nested affine information classes, loss separates into an irreducible information energy and an additional implementation energy. These are conditional statements about a fixed quadratic objective and exogenous law. With known impact, causal integration and projection specify the forecast-error metric exactly. Constant bias does not maximize that metric in general. In the stationary tracking specialization, wrong persistence and wrong friction have explicit excess costs, and the local expected persistence loss depends on estimation MSE.

The order-book certificate retains event timing, price marks, hidden-state inference, and state reduction. The event filter computes a posterior under a declared parameter model; it does not identify those parameters from data. Its perturbation bound can deteriorate on rare histories. The coarse-state certificate includes both transition-rate and price-compensator residuals, because transition lumpability alone does not preserve marked price changes. The additive forecast's three horizon coefficients are individually sharp. Positive likelihood and future-increment enclosures can be

substantially more informative than a succession of norm bounds. Their validity requires a positive history-likelihood lower bound, rigorous arithmetic enclosures, and an uncertainty set containing the true primitives. Their interval minimax claim applies to that interval, which can enlarge a correlated parameter family. A price-specific feature can preserve every conditional mean without preserving the full transition law. This conclusion requires the stated common friction and price compensator; a state-dependent risk charge or fill objective can need additional observables. The information decomposition also separates unobserved current-state variation from model or implementation error. In the delayed-feed model the former is generally linear in small latency, even when the latter has a quadratic forecast loss. The stationary latency frontier and its no-trade threshold concern the specified observation experiment, with no additional informative events between snapshots. The OU result uses a fully observed, exogenous, stationary scalar target and excludes terminal constraints and feedback of trading into the target. Its excess-cost identity evaluates tracking rules within that model. Extending these certificates to calibrated venue-specific books, nonstationary latent processes, and endogenous impact requires additional assumptions and analysis.

The accompanying checks verify exact formulas against independent optimization, matrix-exponential, covariance, and finite-tree adjoint calculations. Twenty-four random affine and mixed-constraint instances check the joint residual; the largest affine-identity discrepancy is below 10^{-15} . Exact rational calculations reproduce Example 6.5; numerical integrations recover the sharp threshold exponents for $\alpha = 1/2, 1, 2$. Additional checks verify the curvature–diameter envelope in 192 independent scalar optimizations, with rational examples attaining both branches. Quadrature and a singular-value computation verify the sharp causal integration constant. Ninety-six independent stationary Lyapunov calculations check the persistence and friction formulas, with maximum discrepancy below 8×10^{-15} . The event filter is independently compared with its normalized nonlinear ODE, and the aggregation identity with matrix-exponential quadrature. Example 5.12 checks the complete transfer from an observed event history through a reduced model to an inexact position decision. Timing and state-reduction counterexamples show why the declared observation and state variables matter. The additional interval calculations use rational arithmetic to enclose exponentials, future-price integrals, and normalized forecasts. Independent optimization checks the minimax rule for 40 mixed-friction cases and three nonquadratic frictions; exact examples attain its quadratic and threshold radii. These computations distinguish numerical comparisons from the analytic inequalities and outward rounding that certify a whole uncertainty box. These are synthetic verification exercises, not a comparison of neural architectures or a backtest of market predictability.

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